

salmon outmigrant phenology

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introduction

The intent here is to explore the beach seine data from the DJFMP dataset for patterns in the presence and abundance of juvenile Chinook salmon in the lower Sacramento River and the Delta. These data were first collected in 1976 and the most recent data are from water year 2024.

setup

One package *not* available on CRAN: If necessary, get it using `remotes::install_github("doi-usgs/EflowStats@v5.2.0")`

```
# library(conflicted) # causes WAAAY more trouble than it solves!
library(tidyverse)
library(here)
library(readxl)
library(janitor)
library(ggplot2)
library(MASS)
library(brms)
library(chron)
library(gridExtra)
library(dataRetrieval)
library(EflowStats)
library(fishualize)
library(tidybayes)
```

load data

```
# beach seine data
raw <-
  read_rds(here("salmon/data/raw/raw.rds"))

# naming conventions
taxonomy <-
  read_rds(
    here("salmon/data/raw/dt4.rds"))

# site information
sample_sites <-
  read_rds(
```

```

here("salmon/data/raw/dt5.rds"))

# pull all IEP fish codes out to help with filtering
# for fish data only (ie no inverts)
fish_codes <-
  taxonomy %>%
  filter(Phylum == "Chordata",
         # remove records where Species is NA
         # but retains eg "Bass unidentified"
         Species != "<NA>") %>%
  distinct(IEPFishCode)

```

wrangle data

choose sites

There are beach seine data from a large number of sites, but the consistency with which these sites were sampled varies widely. In a separate document, I provide the details used to select sites that were sampled consistently from water year 1976-2024. Even at these sites, however, there are missing data, particularly for the months April-December prior to 1992.

```

# list of beach seine station codes identified minimally adequate for
# temporal coverage
good <-
  tibble(station = read_rds(here("salmon/data/processed/selected_site_codes.rds")))

# select data from those minimum effort sites and for complete water years (1976-10-01 onwards)
dat <-
  raw %>%
  filter(station %in% good$station,
         # limit to data collected during complete water years
         date >= "1976-10-01") %>%
  mutate(year = year(date),
         month = month(date)) %>%
  relocate(region, .after = id_code) %>%
  relocate(year:month, .before = date)

reg_labs <-
  tibble(region = factor(1:6),
         region_label = c("Lower Sac R", "N Delta", "Cen Delta", "S Delta",
                          "Lower SJR", "SF & SP Bays"))

# create monthly coverage indicator and retain region_code
coverage <- dat %>%
  distinct(station, year, month, region) %>%
  # 'tag' months sampled
  mutate(sampled = 1)

```

```
# create tibble of station_code and region_code for later use
sta_reg <-
  coverage %>%
  distinct(station, region) %>%
  left_join(reg_labs, by = "region")
```

water year types

In California, *water year types* are classifications (Wet, Above Normal, Below Normal, Dry, and Critical) that are used to categorize hydrological conditions over a 12-month period (October 1–September 30), based, for the Sacramento and San Joaquin valleys, on unimpaired runoff. These designations are used for making water management decisions and probably are not the best environmental measures available for understanding salmon ecology, but they are used by managers and offer a reasonable starting point for exploring the relationship between environment and salmon migration. Additional environmental metrics will be explored in the future.

Add data on *water year types* to the data:

```
# add variables: water year and day of water year
# select data beginning with WY 1976
# (Oct 1 1976 - Sep 30 1977) and ending with WY 2024 (Oct 1 2024 - end of
# data set (July 31, close enough to the proper end of the WY for our purposes))

dat <-
  dat %>%
  # add water year (wy) and day of water year (wd) variables
  mutate(
    # water year
    wy = dataRetrieval::calcWaterYear(date),
    # day of water year (wd), wyMonth=10L bc water year begins Oct 1
    wd = EflowStats::get_waterYearDay(date, wyMonth = 10L)) %>%
  # reorg & select columns
  relocate(wy:wd, .after = date) %>%
  dplyr::select(c(station:water_temp, organism_code, fork_length, race_by_length, count)) %>%
  # add 'sample_code' to identify each unique seine event
  group_by(date, time, station) %>%
  mutate(sample_id = cur_group_id()) %>%
  ungroup() %>%
  relocate(sample_id)
```

analysis

accumulation curves

Calculate the accumulated number of salmon over the course of each water year and aggregate over selected beach seine sites. Determine a cumulative percentage for each day over the entire year and for the 3-month period January, February, and March. Plot annual accumulation curves to show the pattern of juvenile outmigration for each water year, as well as for each January-March period for each year. The truncated data are used as these months were sampled consistently throughout the time series; sampling effort during the remaining months, April-December was limited prior to 1992.

```
# daily counts
# pivot to simplify summing Chinook counts (including 0's) for each sample unit
seine <-
  dat %>%
  pivot_wider(
    # relies on 'sample_id' to distinguish among sample units, but retains environmental data
    id_cols = c(sample_id:time, water_temp),
    # columns for all fish spp (only CHN needed for this analysis)
    names_from = organism_code,
    # log number of each speciesxfork_length for each seine (sample_id)
    values_from = count,
    # if no matches, enter '0'
    values_fill = 0,
    # sum counts if multiple matches
    values_fn = sum
  ) %>%
  dplyr::select(c(sample_id:water_temp, CHN))

### daily counts -----
# aggregate to daily totals bc freq >>1 seine per day
# note that this will obscure potential differences among locations

daily <-
  seine %>%
  group_by(wy, month, wd) %>%
  summarise(
    daily_total = sum(CHN, na.rm = TRUE),
    .groups = "drop"
  )

### cumulative percent-----
# calc cumulative percentage for each wy
daily_cum <-
  daily %>%
  group_by(wy) %>%
  arrange(wy, wd) %>%
  mutate(
    annual_total = sum(daily_total),
    cum_total = cumsum(daily_total),
    cum_percent = 100 * cum_total / annual_total
  ) %>%
  arrange(wy, month, wd) %>%
```

```

ungroup()

# calc cumulative percentage for each wy but limited to the period from Jan 1-Mar 31
# Jan 1 is wd 93; Mar 31 is wd 182 (leap years, it's 183 but I'm going to ignore this)

daily_cum_limited <-
  daily %>%
  filter(month <= 3) %>%
  group_by(wy) %>%
  arrange(wy, wd) %>%
  mutate(
    annual_total = sum(daily_total),
    cum_total = cumsum(daily_total),
    cum_percent = 100 * cum_total / annual_total
  ) %>%
  arrange(wy, month, wd) %>%
  ungroup()

```

```

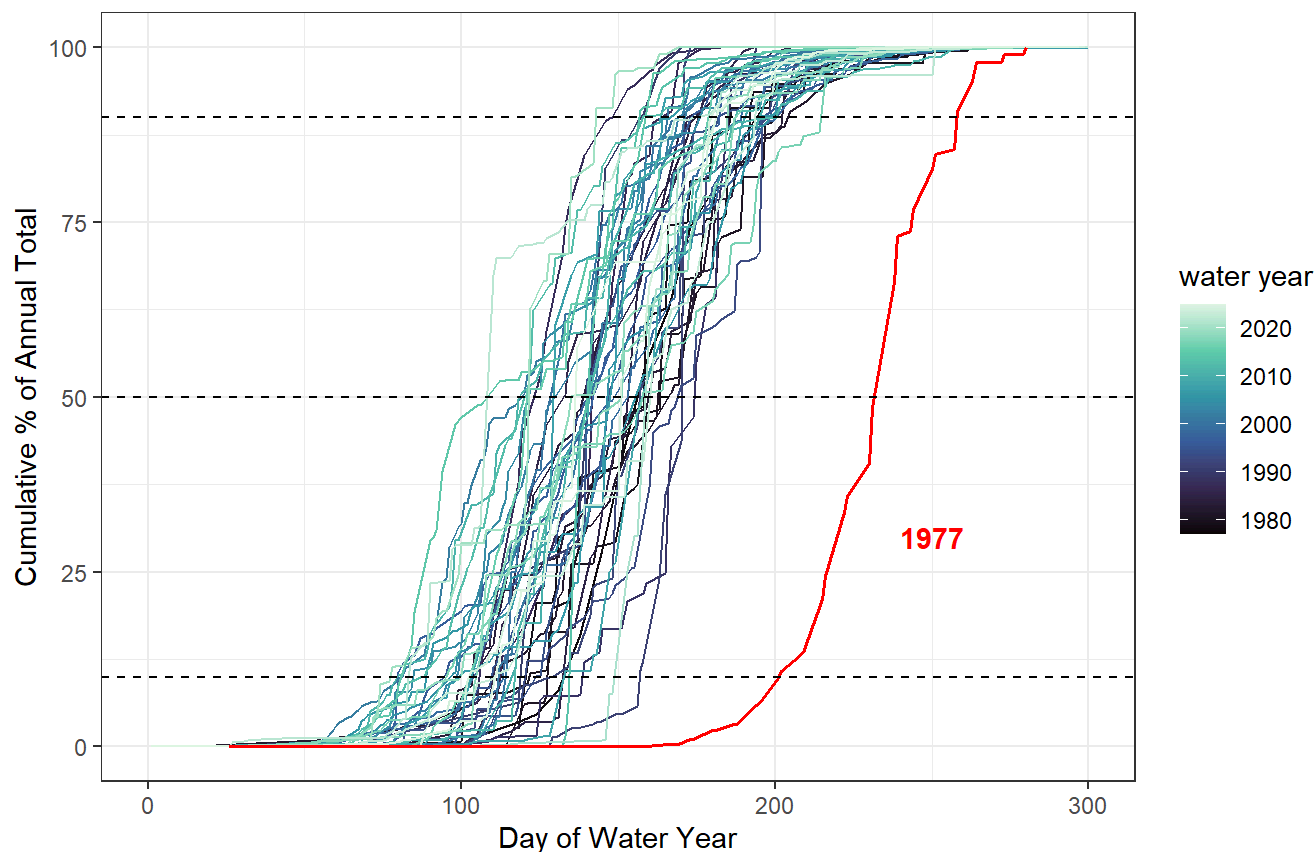
daily_cum %>%
  arrange(wy, wd) %>%
  ggplot(aes(x = wd,
             y = cum_percent,
             group = wy,
             color = wy)) +
  geom_line() +
  # add 1977 on top in red
  geom_line(data = subset(daily_cum, wy == 1977),
            color = "red",
            linewidth = 0.7) +
  xlim(0, 300) +
  geom_hline(yintercept = c(10, 50, 90),
            linetype = "dashed") +
  scale_color_viridis_c(option = "mako") +
  # highlight 1977 data as an apparent outlier
  annotate("text", x = 250, y = 30,
          label = "1977", color = "red", fontface = "bold") +
  labs(x = "Day of Water Year",
       y = "Cumulative % of Annual Total",
       title = "Juvenile Chinook Accumulation Curves",
       subtitle = "data from January - December, 1977-2024",
       color = "water year") +
  theme_bw()

```

Warning: Removed 926 rows containing missing values or values outside the scale range (`geom\_line()`).

Juvenile Chinook Accumulation Curves

data from January - December, 1977-2024

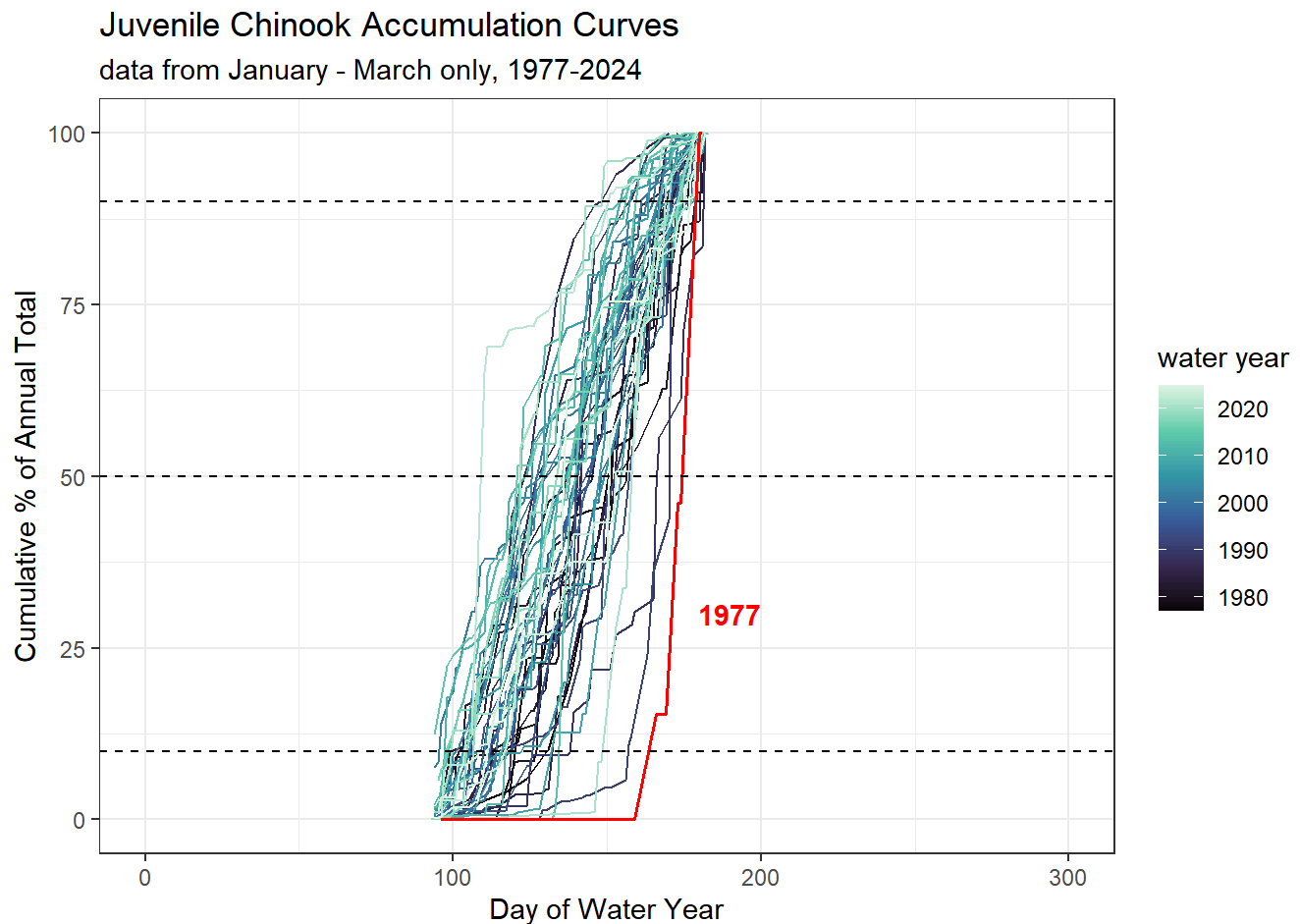


```
ggsave(here("salmon/plots/accum_allmonths.png"))
```

Warning: Removed 926 rows containing missing values or values outside the scale range (`geom\_line()`).

```
# THIS is the approach that we prob want to use
# limit the data to January - March: data = daily_cum_limited
ggplot(daily_cum_limited,
  aes(x = wd,
    y = cum_percent,
    group = wy,
    color = wy)) +
  geom_line() +
  # add 1977 on top in red
  geom_line(data = subset(daily_cum_limited, wy == 1977),
    color = "red",
    linewidth = 0.7) +
  xlim(0, 300) +
  geom_hline(yintercept = c(10, 50, 90),
    linetype = "dashed") +
  scale_color_viridis_c(option = "mako") +
  annotate("text", x = 190, y = 30,
    label = "1977", color = "red", fontface = "bold") +
```

```
labs(x = "Day of Water Year",
     y = "Cumulative % of Annual Total",
     title = "Juvenile Chinook Accumulation Curves",
     subtitle = "data from January - March only, 1977-2024",
     color = "water year") +
theme_bw()
```



```
ggsave(here("salmon/plots/accum_jan2mar.png"))
```

benchmarks

Determine the day of each water year when 10%, 50% and 90% of the total annual numbers are reached.

```
## ID benchmarks ----
# day of water year when the 10%, 50% and 90% of total annual numbers--accumulation benchmarks

benchmarks <- c(10, 50, 90)

# gen df of water year days when threshold reached for each year
percentile_days <-
  daily_cum %>%
```

```

group_by(wy) %>%
summarise(
  day_10 = min(wd[cum_percent >= 10]),
  day_50 = min(wd[cum_percent >= 50]),
  day_90 = min(wd[cum_percent >= 90]),
  .groups = "drop"
)

# gen df of water year days when threshold reach for each year, using data from
# January - March only
percentile_days_limited <-
  daily_cum_limited %>%
  group_by(wy) %>%
  summarise(
    day_10 = min(wd[cum_percent >= 10]),
    day_50 = min(wd[cum_percent >= 50]),
    day_90 = min(wd[cum_percent >= 90]),
    .groups = "drop"
  )

# pivot data to long format to facilitate graphics
pctl_days <-
  percentile_days %>%
  pivot_longer(cols = c(day_10, day_50, day_90),
               names_to = "percentile",
               values_to = "wd") %>%
  mutate(percentile = factor(percentile))

pctl_days_limited <-
  percentile_days_limited %>%
  pivot_longer(cols = c(day_10, day_50, day_90),
               names_to = "percentile",
               values_to = "wd") %>%
  mutate(percentile = factor(percentile))

```

water year type

Add the water year type data (loaded previously) to the percentile data for analysis and graphics.

```

## year type ----
# first, add water year type to the percentile_days df
wy_type <-
  read_excel(here("salmon/data/raw/cdec-water-year-type-jun-2025.xlsx")) %>%
  mutate(wy = WY,
         wyt = factor(WYT),
         watershed = factor(Area),
         .keep = "unused")

pctl_days_wyt <-

```



```

wy_type %>%
# select Sac watershed, excludes SJR watershed
filter(watershed == "Sacramento Valley") %>%
# merge w simple df, includes 1977 data
right_join(pctl_days,
            by = "wy") %>%
dplyr::select(-watershed) %>%
# specify order of the wyt levels
mutate(wyt = fct_relevel(wyt, "W", "AN", "BN", "D", "C"),
        # center wy to facilitate model convergence
        wy_centered = wy - 1977) %>%
relocate(wy_centered, .after = wy)

# make 'wyt' an ordered factor
pctl_days_wyt <- pctl_days_wyt %>%
  mutate(wyt = factor(wyt, levels = c("C", "D", "BN", "AN", "W"),
                      # must be low to high, where C<D<BN<AN<W
                      ordered = TRUE))

write_rds(pctl_days_wyt,
          here("salmon/data/processed/pctl_days_wyt.rds"))

```

migration timing

Plot the water year days when each benchmark (10, 50, or 90%) is reach for each water year.

```

# figure shows the day of the water year for each benchmark (10, 50, 90%) for each year

benchmark_trends_limited <-
  pctl_days_limited %>%
  ggplot(
    # color points according to threshold (10, 50, 90%)
    aes(x = wy, y = wd, color = percentile)
  ) +
  geom_point(size = 4) +
  scale_color_fish(discrete = TRUE,
                  option = "Oncorhynchus_tshawytscha",
                  alpha = 0.6) +
  geom_smooth(data = pctl_days_limited[4:147,],
             alpha = 0.2,
             # show linear trends
             method = "lm",
             se = TRUE) +
  labs(title = "Chinook Salmon Outmigration Timing",
       subtitle = "data from January-March only; 1977 excluded from trend analyses",
       x = "Water Year",
       y = "Day of the Water Year",
       color = "Percentile") +
  theme_bw()

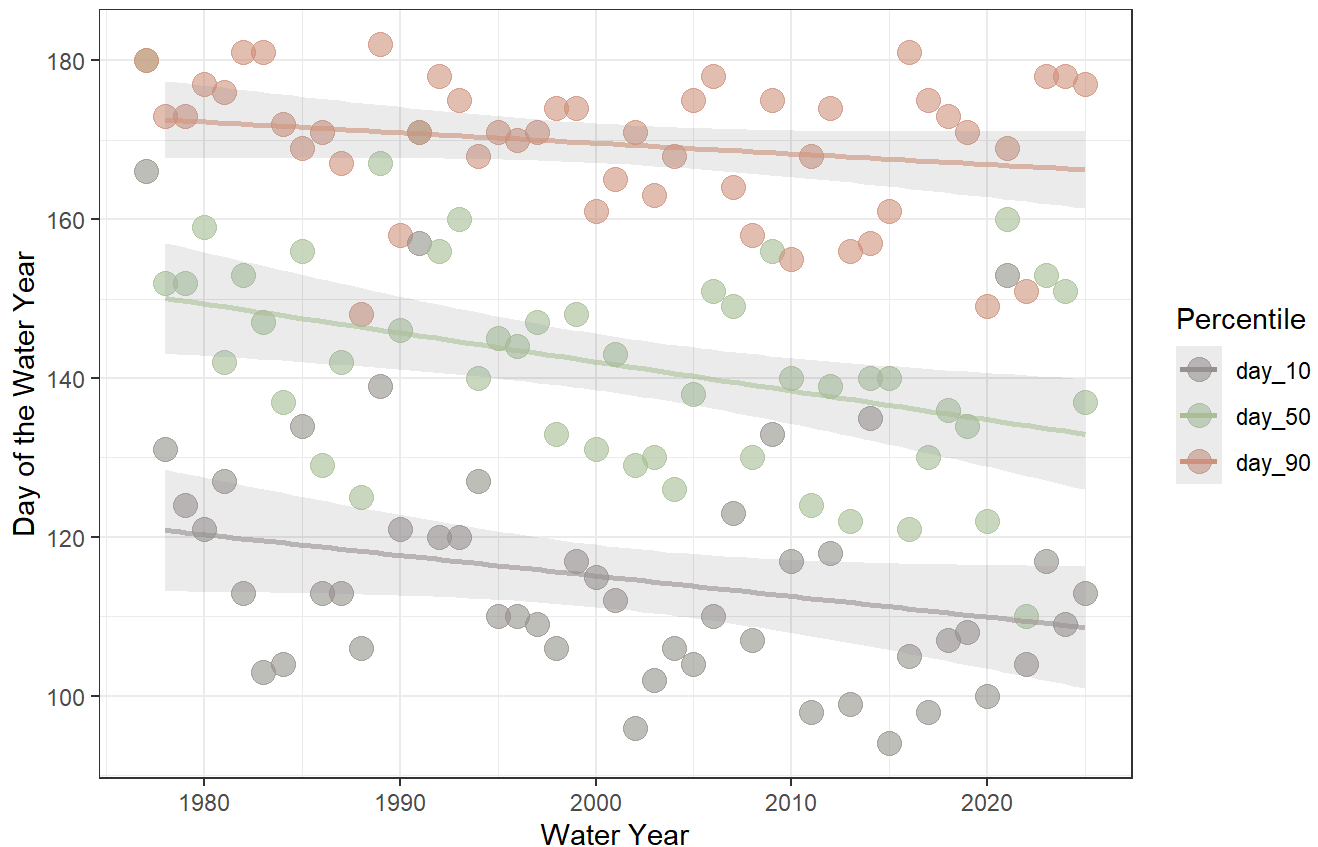
```

benchmark_trends_limited

``geom_smooth()` using formula = 'y ~ x'`

Chinook Salmon Outmigration Timing

data from January-March only; 1977 excluded from trend analyses



models

year effect

Testing for a water year effect, focusing first on the *day10* benchmark, when 10% of each annual total has been reached. Code includes multiple checks on model efficacy, and comparisons between alternative model structures.

```
# first including data from 1977
# reaching the 10% benchmark
m3.day10.wy <-
  brm(data = percentile_days_limited, # use the percentile_days_limited data
    # use a gaussian (normal) distribution
    family= gaussian,
    # specify the model
    day_10 ~ 1 + wy,
    # specify parameters for executing the Markov chains
```

```

iter = 2000, warmup = 1000, chains = 4, cores = 4,
# set seed
seed = 42,
file = here("salmon/results/m3.day10.wy"),
file_refit = "on_change")

```

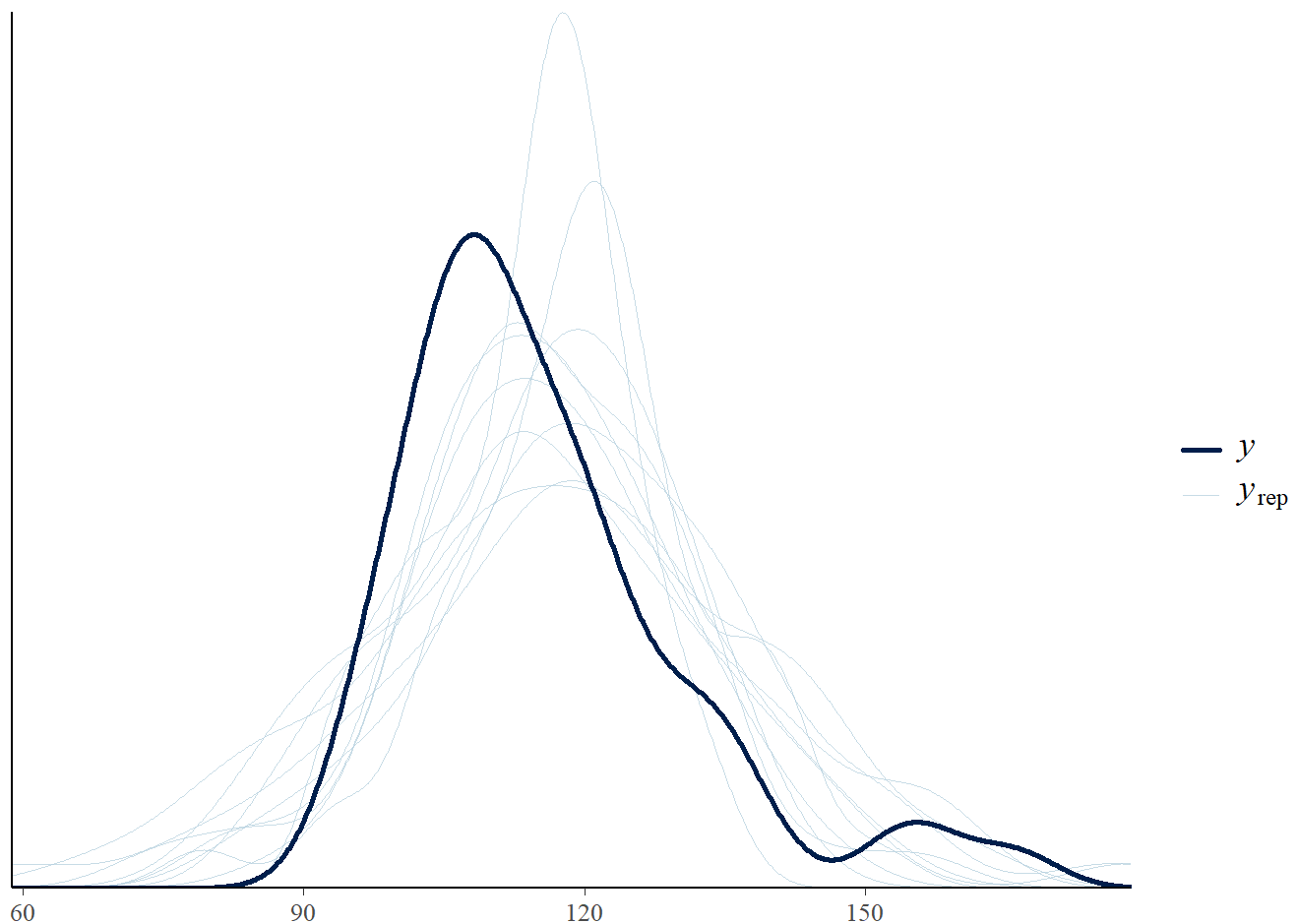
```
write_rds(m3.day10.wy, here("salmon/results/m3.day10.wy.rds"))
```

```

# run posterior predictive check
pp_check(m3.day10.wy)

```

Using 10 posterior draws for ppc type 'dens_overlay' by default.



```

# calc the prob that the slope is less than 0

m3.day10.wy %>%
  spread_draws(b_wy) %>%
  summarise(p_slope_lessthan_zero = sum(b_wy < 0) / n())

```

```

# A tibble: 1 × 1
  p_slope_lessthan_zero

```

<dbl>

1

0.995

```

m3.day10.wy_robust <-
  brm(data = percentile_days_limited,
    family = student, # <--- change from gaussian to student
    day_10 ~ 1 + wy,
    # apply loo_moment_match on problematic observations bc crazy 1977 outlier
    save_pars = save_pars(all = TRUE),
    iter = 2000, warmup = 1000, chains = 4, cores = 4,
    seed = 42,
    file = here("salmon/results/m3.day10.wy_robust"),
    file_refit = "on_change")

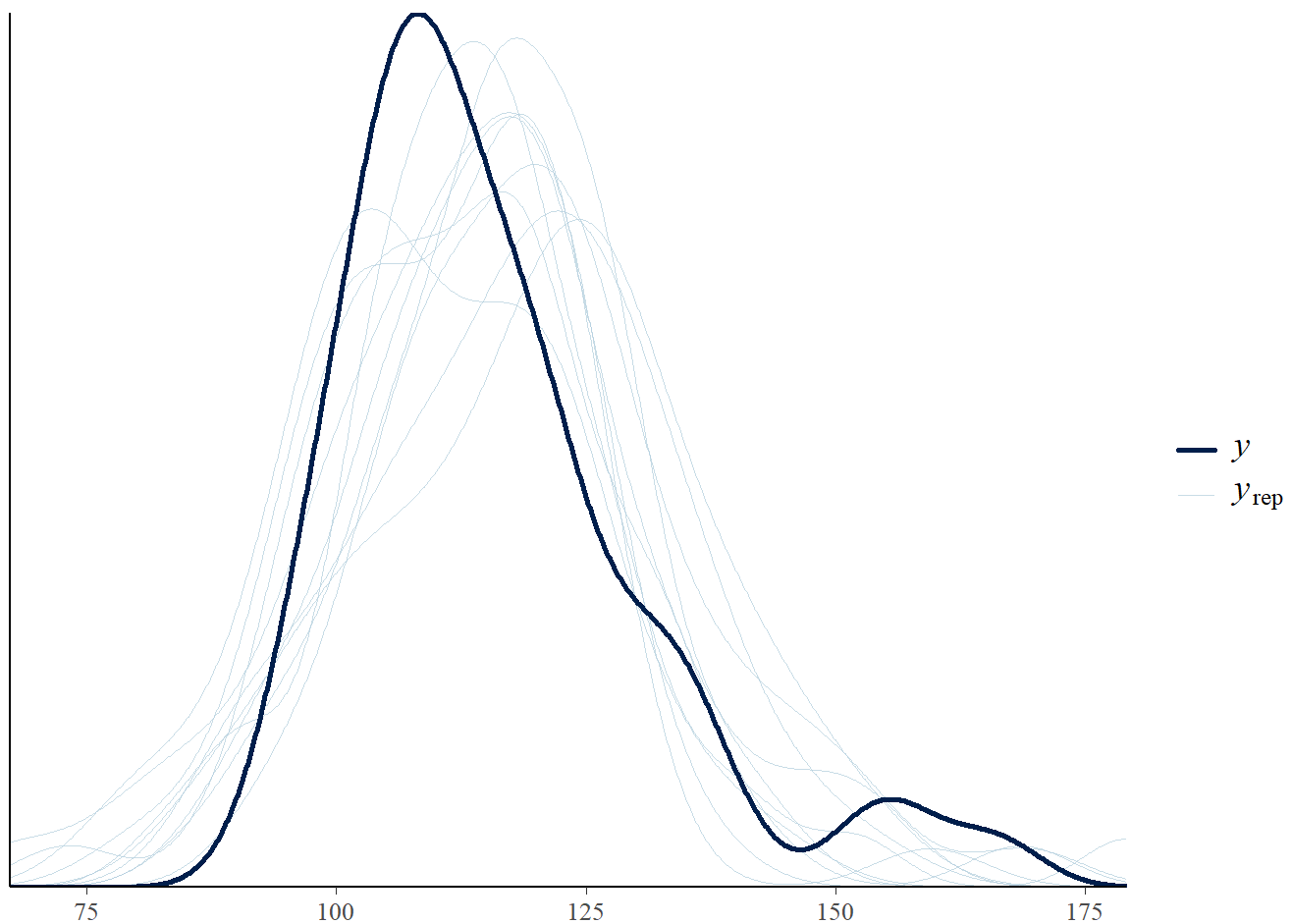
```

```

# looks a bit better!
pp_check(m3.day10.wy_robust)

```

Using 10 posterior draws for ppc type 'dens_overlay' by default.



```

# calc the prob that the slope is less than 0

```

```

m3.day10.wy_robust %>%

```

```
spread_draws(b_wy) %>%
summarise(p_slope_lessthan_zero = sum(b_wy < 0) / n())
```

```
# A tibble: 1 × 1
  p_slope_lessthan_zero
      <dbl>
1             0.996
```

```
# calculate LOO for the original Jan-Mar Gaussian model
m3.day10.wy <- add_criterion(m3.day10.wy, criterion = "loo")
# warning about "Pareto values > 0.7" happens when one data point (like 1977) is so influential t

# calculate LOO for the new "robust" (Student-t) model
m3.day10.wy_robust <- add_criterion(m3.day10.wy_robust, criterion = "loo")

# compare the two day10 models
loo_compare(m3.day10.wy, m3.day10.wy_robust, criterion = "loo")
```

	elpd_diff	se_diff
m3.day10.wy_robust	0.0	0.0
m3.day10.wy	-2.5	1.6

```
# notes:
# the performance difference between the models is very modest:
# the rule of thumb is: consider one model "clearly better" if the elpd_diff is more than twice t

m3.day10.wy_robust$criteria$loo
```

Computed from 4000 by 49 log-likelihood matrix.

	Estimate	SE
elpd_loo	-201.5	7.8
p_loo	4.1	0.9
looic	403.0	15.6

MCSE of elpd_loo is 0.1.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.4, 0.9]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.

```
m3.day10.wy$criteria$loo
```

Computed from 4000 by 49 log-likelihood matrix.

	Estimate	SE
elpd_loo	-204.0	7.7
p_loo	4.1	1.7
looic	408.1	15.5

MCSE of elpd_loo is 0.1.

MCSE and ESS estimates assume MCMC draws (r_{eff} in $[0.7, 1.0]$).

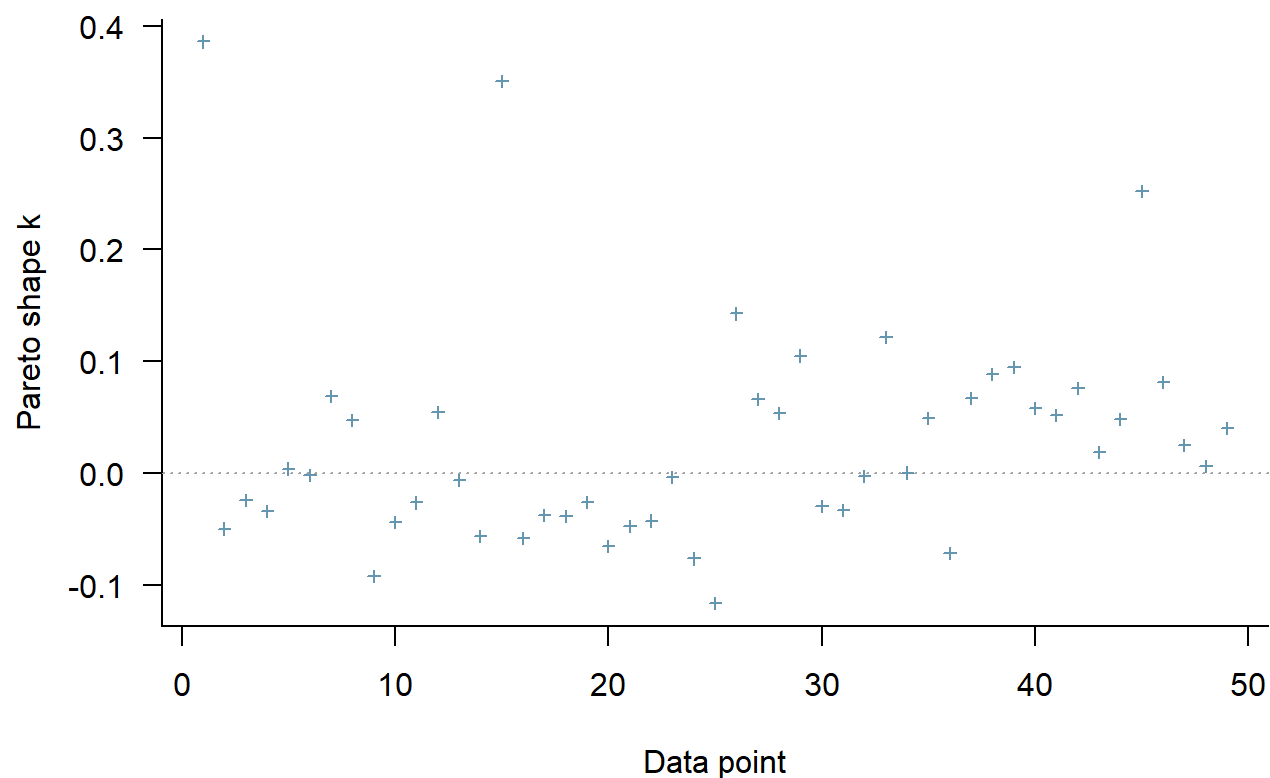
All Pareto k estimates are good ($k < 0.7$).

See `help('pareto-k-diagnostic')` for details.

```
# the robust model (using Student-t distribution) seems to perform slightly better,
# but the improvement could plausibly be attributed to noise
```

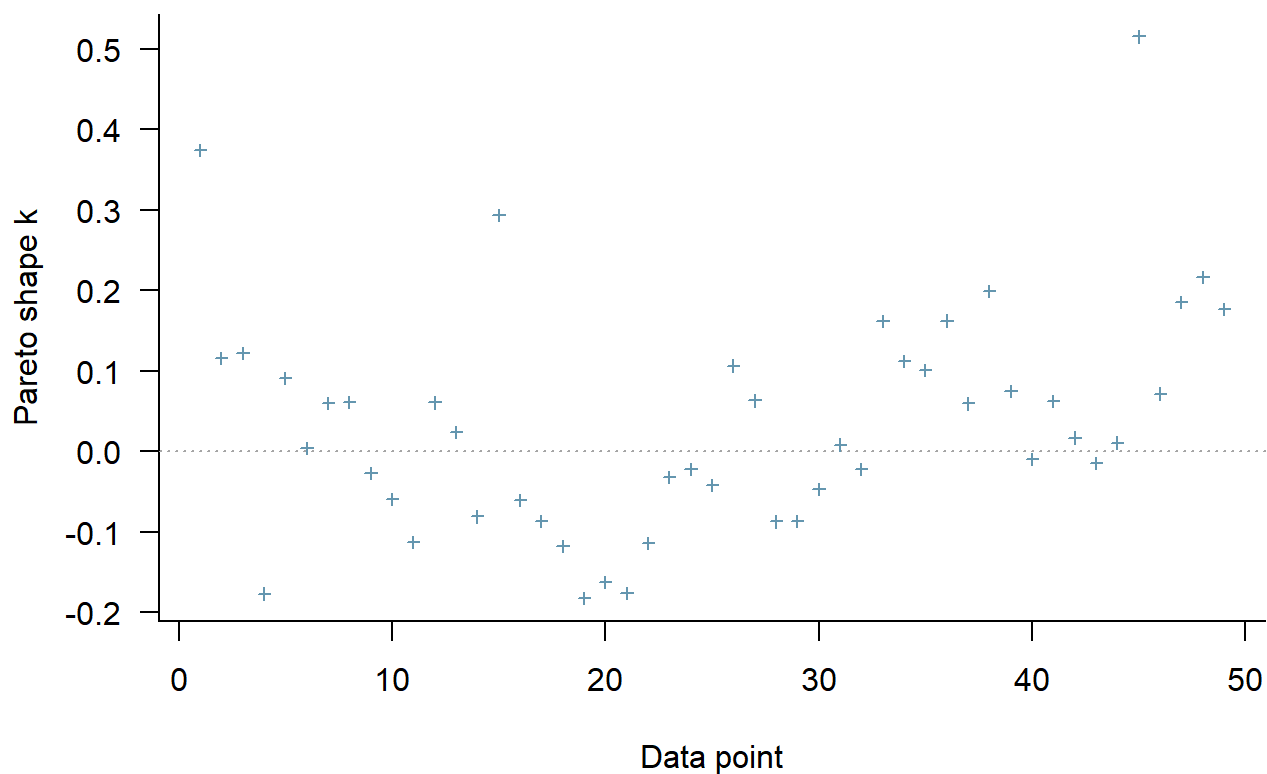
```
# let's plot the k values for each model
plot(m3.day10.wy$criteria$loo)
```

PSIS diagnostic plot



```
plot(m3.day10.wy_robust$criteria$loo)
```

PSIS diagnostic plot



```
# PSIS diagnostic plots confirm that both models are pretty good, but also
# show how nicely the 'robust' model handles the extreme datum from 1977
```

```
summary(m3.day10.wy_robust)
```

```
Family: student
Links: mu = identity
Formula: day_10 ~ 1 + wy
Data: percentile_days_limited (Number of observations: 49)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	821.99	268.03	293.73	1349.03	1.00	3358	2562
wy	-0.35	0.13	-0.62	-0.09	1.00	3361	2511

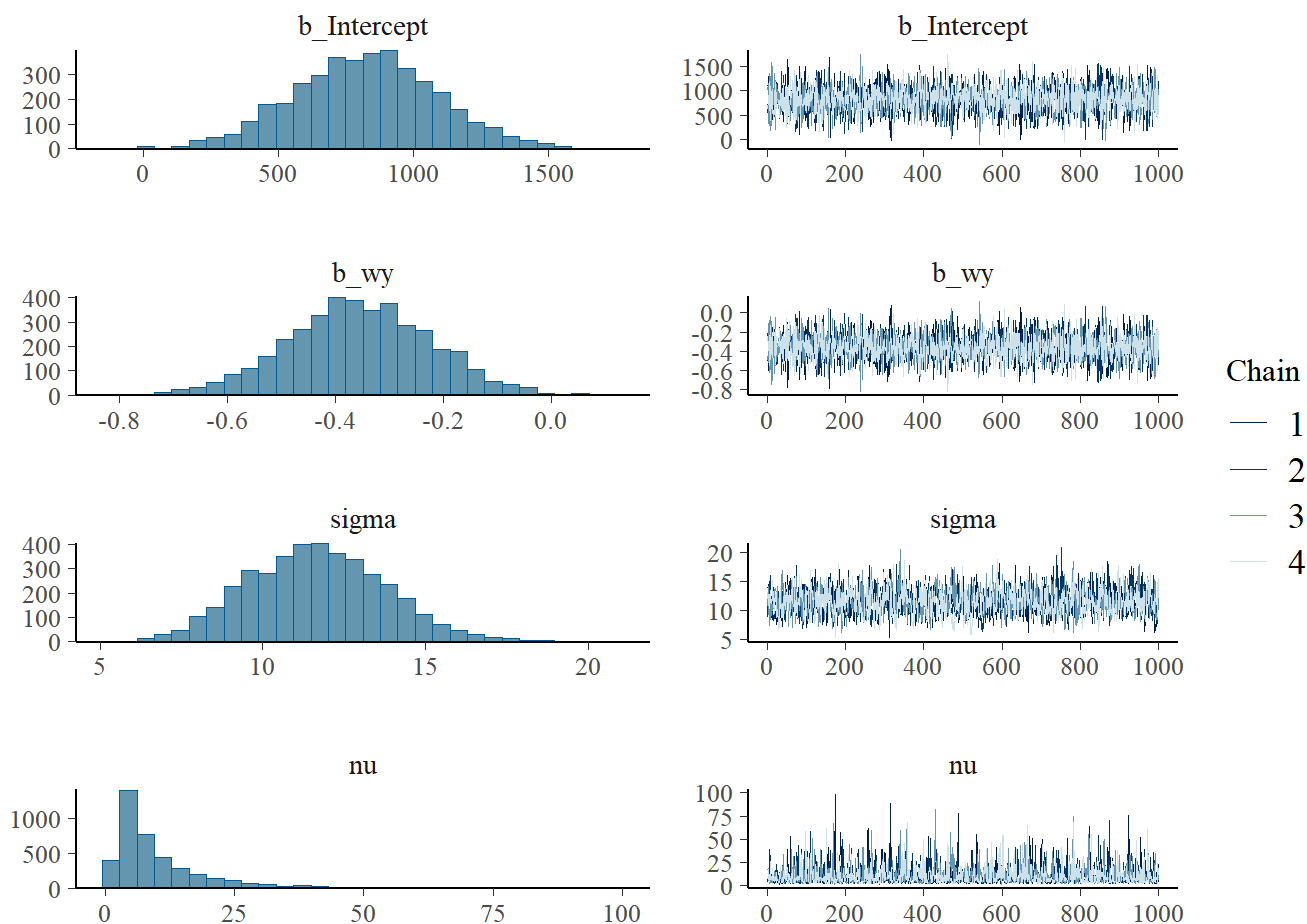
Further Distributional Parameters:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	11.65	2.12	7.80	15.89	1.00	1693	1861
nu	10.25	9.76	1.99	38.34	1.00	1935	2595

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS

and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(m3.day10.wy_robust)
```



```
m3.day10.wy_robust %>%
  spread_draws(b_wy) %>%
  summarise(p_slope_lessthan_zero = sum(b_wy < 0) / n())
```

```
# A tibble: 1 × 1
  p_slope_lessthan_zero
                <dbl>
1                   0.996
```

The “robust” model for the *day10* benchmark provides strong support for the hypothesis that migration, particularly the earlier phase of the process, has shifted earlier in the water year by approximately 0.35 days each year from 1997-2024.

The code below models comparable relationships to the previous code, but uses the *day50* and *day90* benchmarks.


```
# reaching the 50% benchmark
m3.day50.wy <-
  brm(data = percentile_days_limited,
      # choose a gaussian (normal) distribution
      family = gaussian,
      day_50 ~ 1 + wy,
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      seed = 42,
      file = here("salmon/results/m3.day50.wy"),
      file_refit = "on_change")

# reaching the 90% benchmark
m3.day90.wy <-
  brm(data = percentile_days_limited,
      family = gaussian,
      day_90 ~ 1 + wy,
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      seed = 42,
      file = here("salmon/results/m3.day90.wy"),
      file_refit = "on_change")
```

```
summary(m3.day50.wy)
```

```
Family: gaussian
Links: mu = identity
Formula: day_50 ~ 1 + wy
Data: percentile_days_limited (Number of observations: 49)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	1011.29	260.43	506.14	1523.52	1.00	3724	3077
wy	-0.43	0.13	-0.69	-0.18	1.00	3726	3128

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	12.96	1.36	10.64	15.94	1.00	3153	2389

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
summary(m3.day90.wy)
```

```
Family: gaussian
Links: mu = identity
Formula: day_90 ~ 1 + wy
Data: percentile_days_limited (Number of observations: 49)
```

Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

Regression Coefficients:

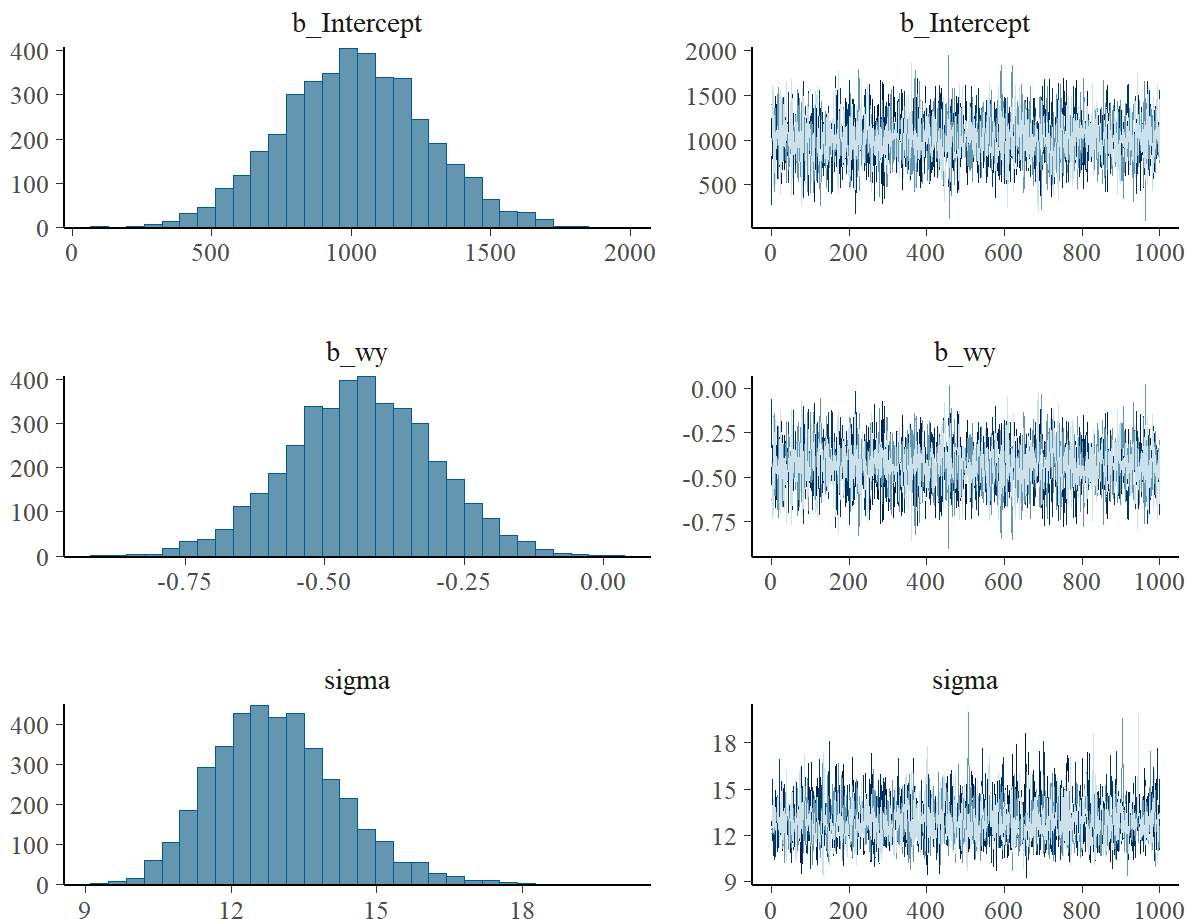
	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	472.91	174.82	138.84	810.15	1.00	3522	2641
wy	-0.15	0.09	-0.32	0.02	1.00	3529	2636

Further Distributional Parameters:

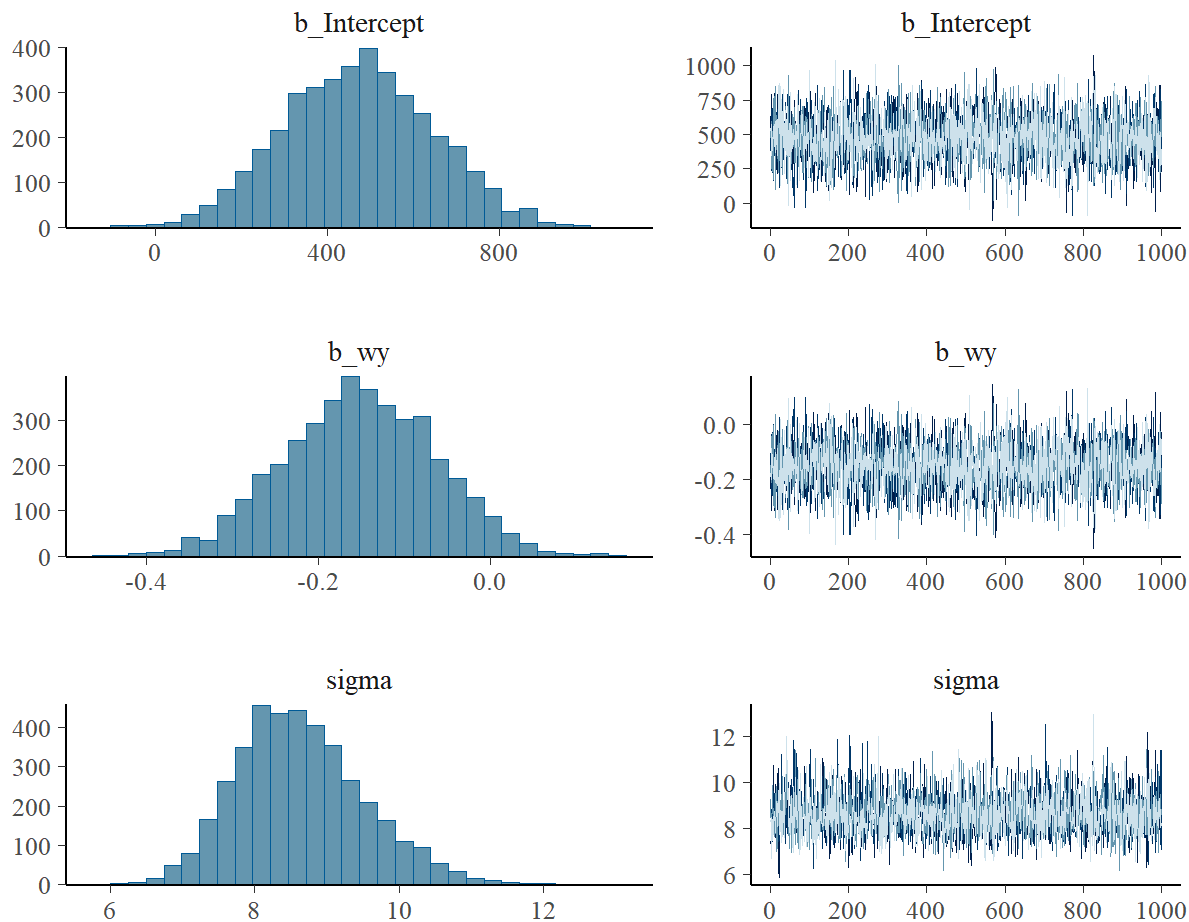
	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	8.64	0.90	7.10	10.55	1.00	3726	3123

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(m3.day50.wy)
```



```
plot(m3.day90.wy)
```



```
# calc the prob that the slope is less than 0
```

```
m3.day50.wy %>%
  spread_draws(b_wy) %>%
  summarise(p_slope_lessthan_zero = sum(b_wy < 0) / n())
```

```
# A tibble: 1 × 1
  p_slope_lessthan_zero
      <dbl>
1             1.000
```

```
m3.day90.wy %>%
  spread_draws(b_wy) %>%
  summarise(p_slope_lessthan_zero = sum(b_wy < 0) / n())
```

```
# A tibble: 1 × 1
  p_slope_lessthan_zero
      <dbl>
1             0.961
```

I tested multiple models to explore the timing of juvenile salmon outmigration, represented by the day of the water year when three benchmarks had been reached over the course of the time series. Each model

used a Gaussian distribution with *water year* as the predictor and the day of the water year when those benchmarks—10, 50, and 90—are reached as the response variable: *day10*, *day50*, *day90*. I limited the data to those gathered during January, February and March because these were sampled consistently (albeit imperfectly) in every year. Early outmigrants entering the Delta in December or earlier would be missed, but presumably the most important early numbers would be reasonably well represented.

These results present strong evidence for a *year* effect (i.e., model slopes were different from zero, however, the posterior distribution for the *day90* model overlaps 0 slightly (95% CIs: -0.73, 0.05).

-For the 10% accumulation model (*day10*), there was an estimated slope of -0.69 days per year and the probability that the slope is less than 0 equals 0.999.

-For the 50% accumulation model (*day50*), there was an estimated slope of -0.70 days per year and the probability that the slope is less than 0 equals 1.000.

-For the 90% accumulation model (*day90*), there was an estimated slope of -0.33 days per year and the probability that the slope is less than 0 equals 0.955.

water year type effect

I also attempted to determine if there was an effect of water year type on the timing of Chinook outmigration. Models included both *water year* and *wyt* (water year type) as explanatory variables.

```
# first, add water year type to the percentile_days df
wy_type <-
  read_excel(here("salmon/data/raw/cdec-water-year-type-jun-2025.xlsx")) %>%
  mutate(wy = WY,
         wyt = factor(WYT),
         watershed = factor(Area),
         .keep = "unused")

pctl_days_wyt <-
  wy_type %>%
  # select Sac watershed, excludes SJR watershed
  filter(watershed == "Sacramento Valley") %>%
  # merge w simple df, includes 1977 data
  right_join(pctl_days,
             by = "wy") %>%
  dplyr::select(-watershed) %>%
  # specify order of the wyt levels
  mutate(wyt = fct_relevel(wyt, "W", "AN", "BN", "D", "C"),
         # center wy to facilitate model convergence
         wy_centered = wy - 1977) %>%
  relocate(wy_centered, .after = wy)

# make 'wyt' an ordered factor
pctl_days_wyt <- pctl_days_wyt %>%
  mutate(wyt = factor(wyt, levels = c("C", "D", "BN", "AN", "W"),
                     # must be low to high, where C<D<BN<AN<W
```

```

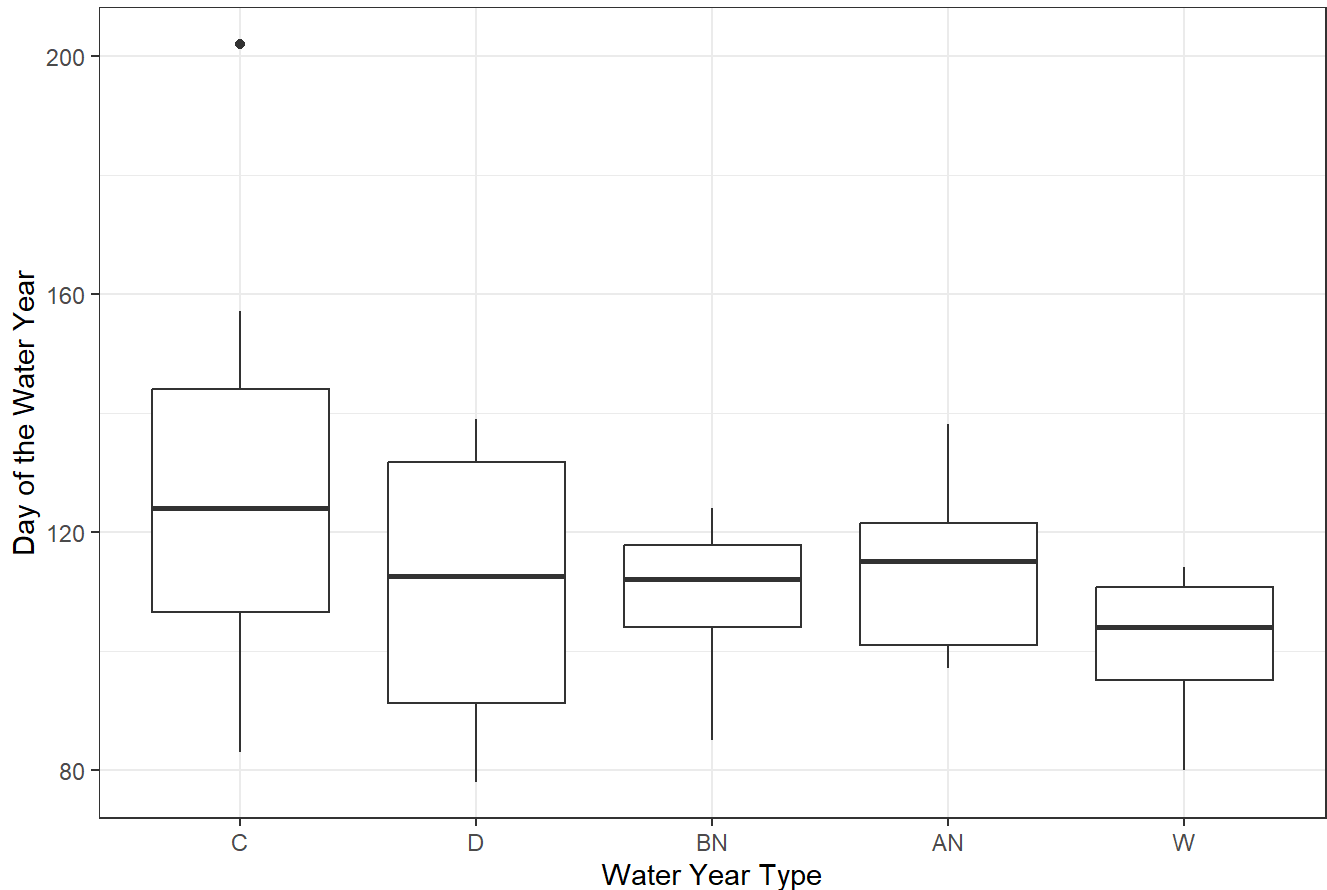
ordered = TRUE))

write_rds(pctl_days_wyt,
          here("salmon/data/processed/pctl_days_wyt.rds"))

### wy boxplot -----
pctl_days_wyt %>%
  filter(percentile == "day_10",
         # no data yet re wyt for 2025
         wy <= 2024) %>%
  ggplot(aes(x = wyt, y = wd, group = wyt)) +
  geom_boxplot() +
  labs(title = "Timing of 10% Annual Accumulation Benchmark",
       x = "Water Year Type",
       y = "Day of the Water Year") +
  theme_bw()

```

Timing of 10% Annual Accumulation Benchmark



```

# run wy + wyt model with the centered variable, where wyt is ordered & monotonic
m1.day10.wy.wyt <-
  brm(data = pctl_days_wyt %>%
       filter(wy <= 2024),
       family = gaussian,
       wd ~ 0 + wy_centered + mo(wyt),

```

```

    iter = 10000, warmup = 1000, chains = 4, cores = 4,
    seed = 42,
    # save fitted model (this one w the wy centered)
    file = here("salmon/results/m1.day10.wy.wyt"),
    file_refit = "on_change"
  )

```

```
summary(m1.day10.wy.wyt)
```

Family: gaussian

Links: mu = identity

Formula: wd ~ 0 + wy_centered + mo(wyt)

Data: pct1_days_wyt %>% filter(wy <= 2024) (Number of observations: 144)

Draws: 4 chains, each with iter = 10000; warmup = 1000; thin = 1;
total post-warmup draws = 36000

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
wy_centered	2.30	0.33	1.65	2.95	1.00	24628	25443
mowyt	25.81	2.91	20.17	31.57	1.00	24609	25234

Monotonic Simplex Parameters:

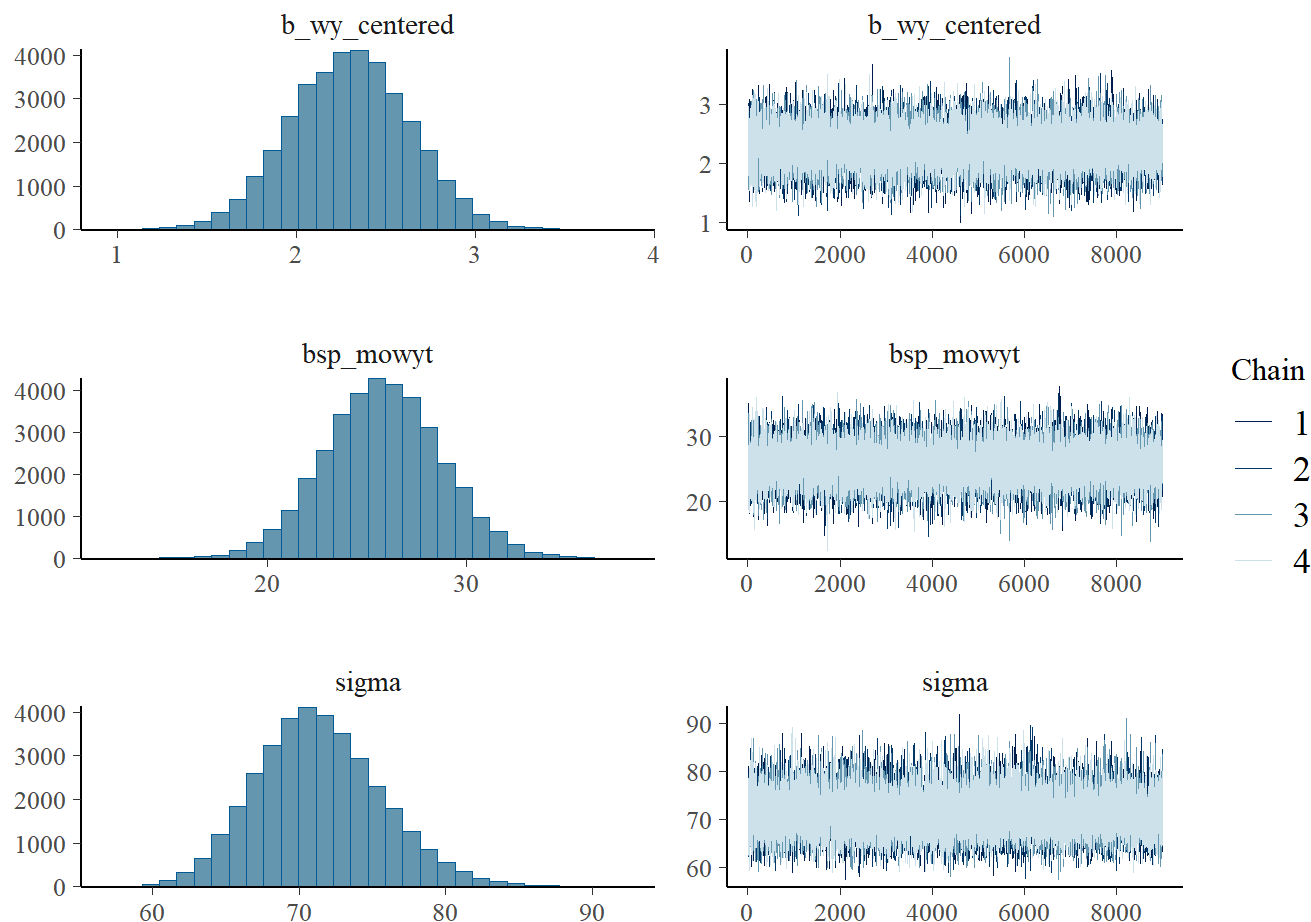
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
mowyt1[1]	0.74	0.10	0.52	0.91	1.00	32080	21164
mowyt1[2]	0.08	0.07	0.00	0.26	1.00	29262	18917
mowyt1[3]	0.10	0.08	0.00	0.29	1.00	32623	23438
mowyt1[4]	0.08	0.06	0.00	0.24	1.00	33496	21927

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	71.38	4.27	63.62	80.38	1.00	39275	26583

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(m1.day10.wy.wyt)
```



```
# Rhat is perfect
# water year (centered) effect looks solid
# posterior distributions look great (uniform)
# mcmc look great too
# estimates for year type are wonky--reflects the data!
# strong effect of 'year type' as a monotonic, ordered factor. cool!
# is there an interaction effect?
# should also include some semi-informed priors?

# compute estimated effect for each wyt level
# get predicted effect of each wyt level
# holding wy_centered at 0
ce <- conditional_effects(m1.day10.wy.wyt, effects = "wyt", re_formula = NA)

# extract the estimate for each level, plus 95% CIs
wyt_effects <-
  ce$wyt[, c("wyt", "estimate__", "lower__", "upper__")]

# round for readability
wyt_effects$estimate <-
  round(wyt_effects$estimate__, 2)

wyt_effects %>%
```

```
dplyr::select(-estimate__) %>%
mutate("l-95% CI" = lower__,
      "u-95% CI" = upper__,
      .keep = "unused")
```

	wyt	estimate	l-95% CI	u-95% CI
1	C	54.10	38.8346	69.31505
2	D	130.15	110.3074	147.68581
3	BN	138.77	120.3685	155.43943
4	AN	149.06	132.4367	166.02756
5	W	157.10	140.7942	174.94971

```
# note that these are scaled effects and can't be translated directly into days of the water year
# Define levels of water year type
levels_wyt <- c("C", "D", "BN", "AN", "W")
```

```
# predicted wd at reference wy_centered = 0
preds <- fitted(m1.day10.wy.wyt, newdata = data.frame(
  wyt = levels_wyt,
  wy_centered = 0
))
```

```
# assemble into a table
predicted_wd <- data.frame(
  wyt = levels_wyt,
  estimate = round(preds[, "Estimate"], 1),
  l_95 = round(preds[, "Q2.5"], 1),
  u_95 = round(preds[, "Q97.5"], 1)
)
```

```
predicted_wd
```

	wyt	estimate	l_95	u_95
1	C	0.0	0.0	0.0
2	D	75.8	51.0	99.4
3	BN	84.4	60.3	107.5
4	AN	95.1	72.2	117.5
5	W	103.2	80.7	126.3

```
# run wy * wyt model with the centered variable
m2.day10.wy.wyt <-
brm(data = pct1_days_wyt %>%
  filter(wy <= 2024),
  family = gaussian,
  wd ~ 0 + wy_centered * wyt,
  iter = 10000, warmup = 1000, chains = 4, cores = 4,
  seed = 42,
  # save fitted model (this one w the wy centered)
```



```
file = here("salmon/results/m2.day10.wy.wyt"),
file_refit = "on_change"
)
```

```
summary(m2.day10.wy.wyt)
```

Family: gaussian
 Links: mu = identity
 Formula: wd ~ 0 + wy_centered * wyt
 Data: pct1_days_wyt %>% filter(wy <= 2024) (Number of observations: 144)
 Draws: 4 chains, each with iter = 10000; warmup = 1000; thin = 1;
 total post-warmup draws = 36000

Regression Coefficients:

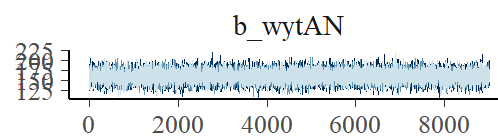
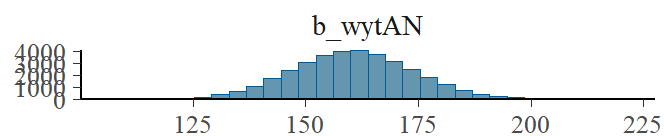
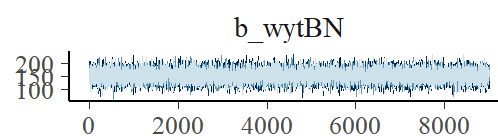
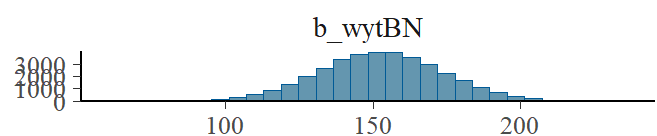
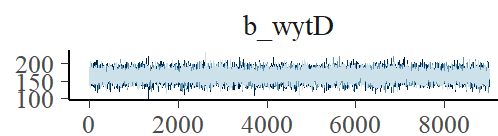
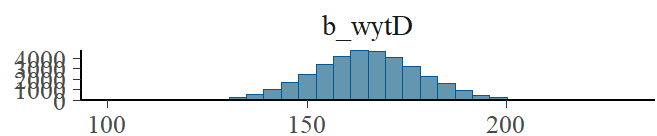
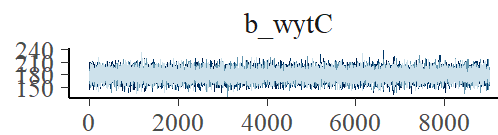
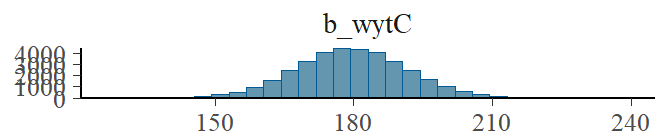
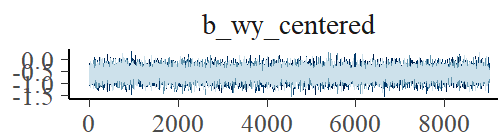
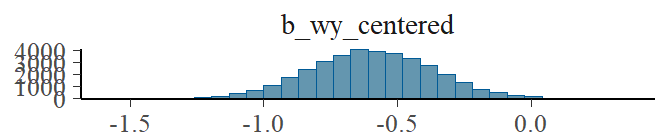
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
wy_centered	-0.60	0.23	-1.05	-0.15	1.00	31314	26992
wytC	179.44	12.03	155.77	203.09	1.00	39237	26614
wytD	164.88	13.40	138.59	191.24	1.00	34538	26687
wytBN	152.61	21.12	111.28	193.80	1.00	26139	24892
wytAN	160.31	13.47	133.89	186.74	1.00	34420	27809
wytW	148.68	10.99	127.15	170.37	1.00	38820	26704
wy_centered:wyt.L	0.85	0.44	-0.02	1.72	1.00	34961	26665
wy_centered:wyt.Q	-0.12	0.52	-1.13	0.91	1.00	30394	26535
wy_centered:wyt.C	0.06	0.51	-0.93	1.06	1.00	33762	26064
wy_centered:wytE4	0.38	0.60	-0.79	1.55	1.00	31748	25612

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	35.88	2.22	31.85	40.56	1.00	47193	26471

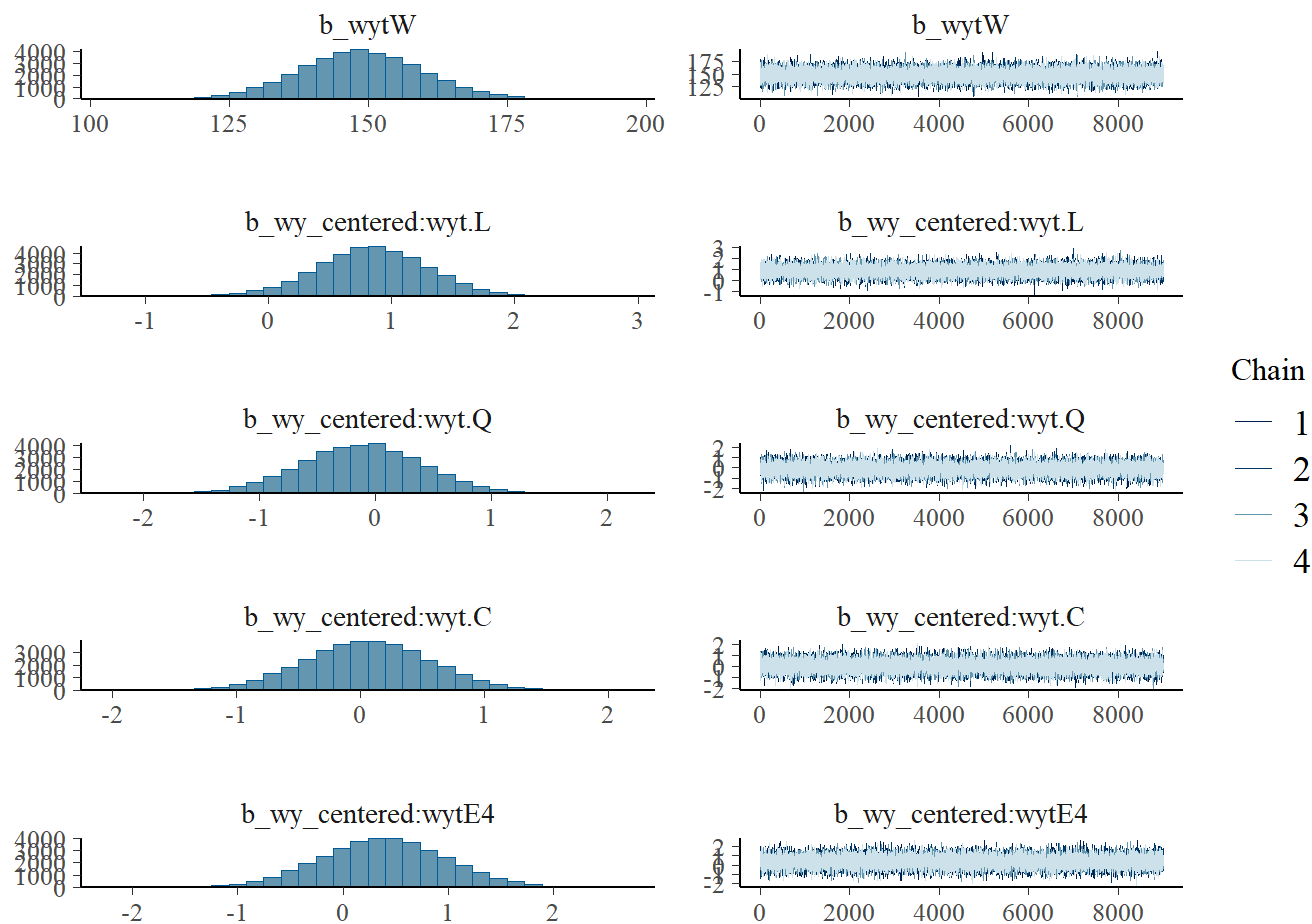
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

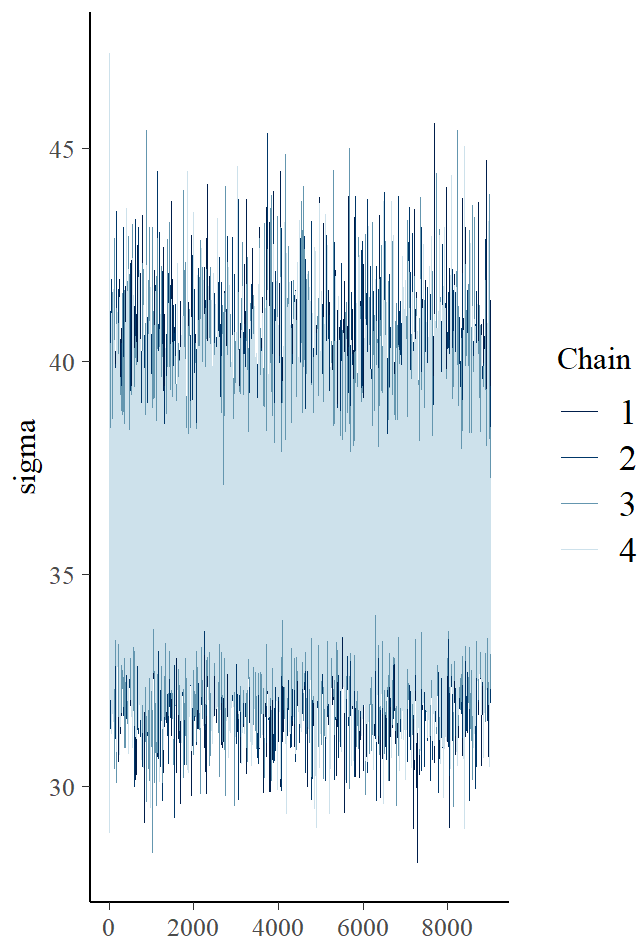
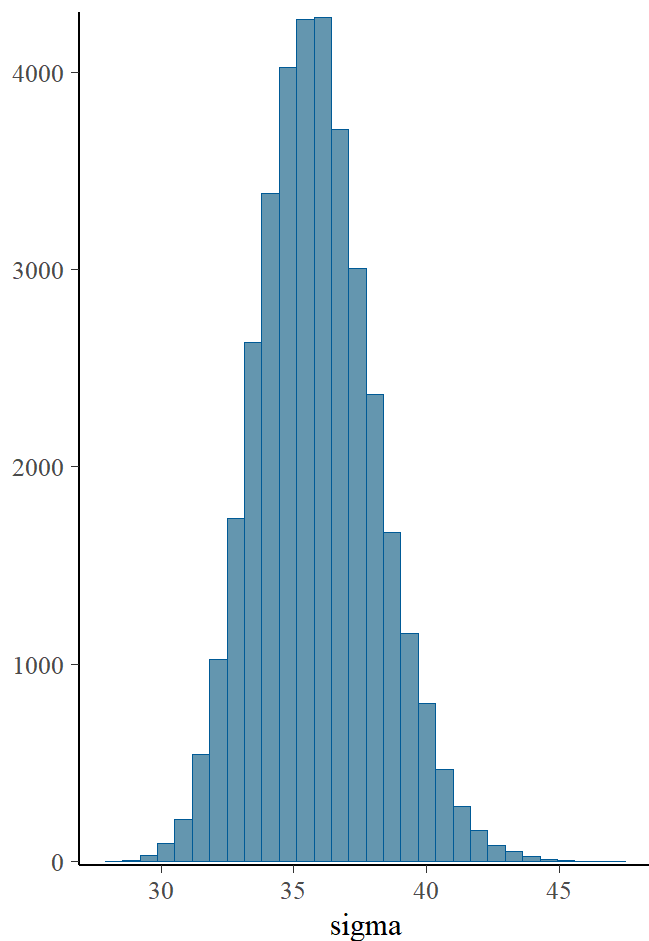
```
plot(m2.day10.wy.wyt)
```



Chain

- 1
- 2
- 3
- 4





```
# run wy * wyt model with the centered variable, 2025 not included, replace Gaussian w Students t
m3.day10.wy.wyt <-
  brm(data = pct1_days_wyt %>%
    filter(wy <= 2024),
    family = student,
    wd ~ 0 + wy_centered * wyt,
    iter = 10000, warmup = 1000, chains = 4, cores = 4,
    seed = 42,
    # save fitted model (this one w the wy centered)
    file = here("salmon/results/m3.day10.wy.wyt"),
    file_refit = "on_change"
  )
```

```
# to be consistent, I should prob designate this model as 'robust', but can't be bothered
summary(m3.day10.wy.wyt)
```

```
Family: student
Links: mu = identity
Formula: wd ~ 0 + wy_centered * wyt
Data: pct1_days_wyt %>% filter(wy <= 2024) (Number of observations: 144)
Draws: 4 chains, each with iter = 10000; warmup = 1000; thin = 1;
```

total post-warmup draws = 36000

Regression Coefficients:

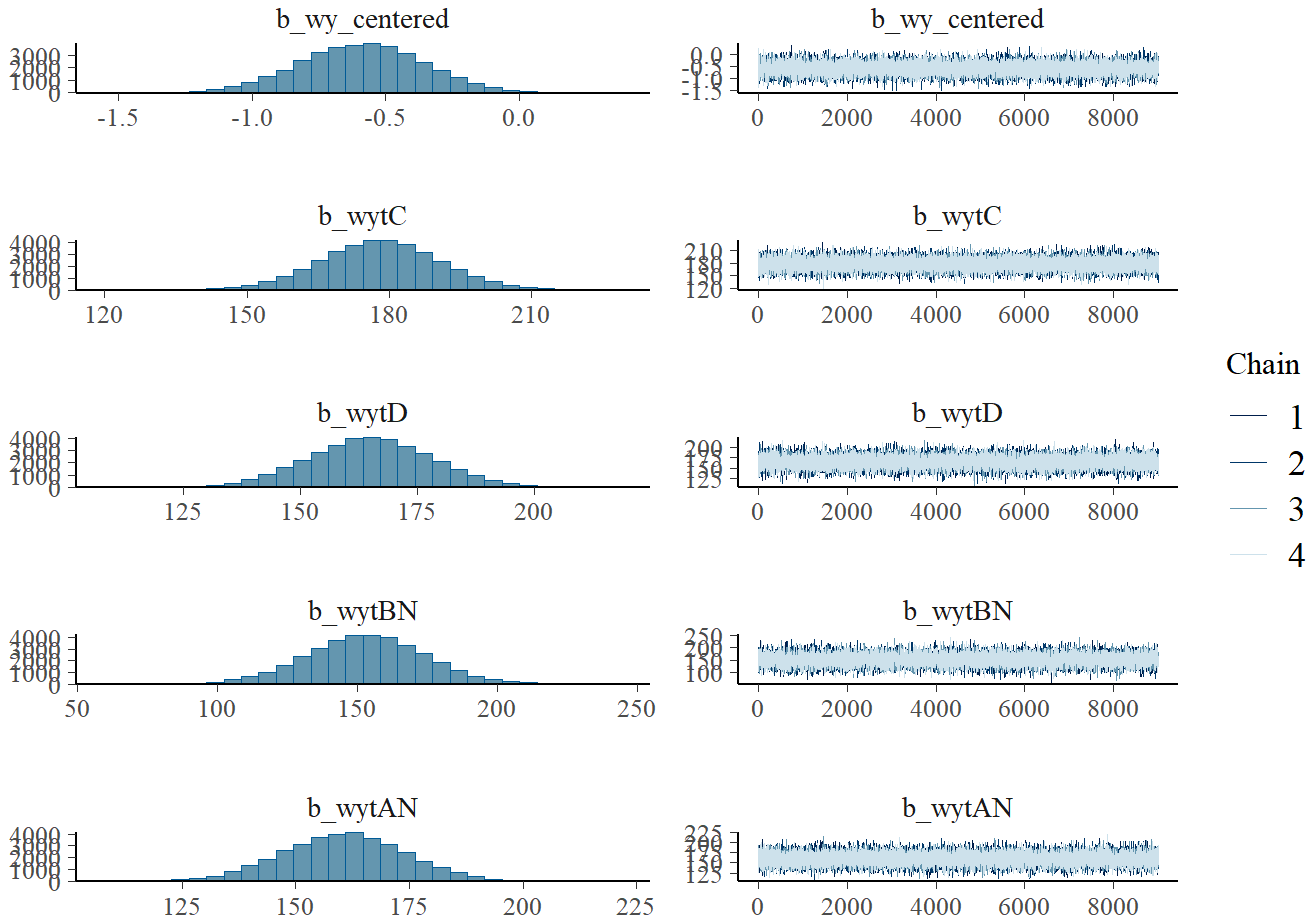
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
wy_centered	-0.59	0.23	-1.05	-0.13	1.00	29984	27020
wytC	178.00	12.58	153.13	202.53	1.00	35587	26155
wytD	164.87	13.25	138.81	190.84	1.00	32621	26740
wytBN	152.88	21.23	110.60	194.63	1.00	26945	25487
wytAN	160.51	13.50	134.01	186.77	1.00	33634	28494
wytw	148.62	11.34	126.03	170.89	1.00	36652	25647
wy_centered:wyt.L	0.82	0.46	-0.08	1.72	1.00	33717	27505
wy_centered:wyt.Q	-0.09	0.53	-1.13	0.95	1.00	29790	25251
wy_centered:wyt.C	0.05	0.51	-0.96	1.07	1.00	32855	27682
wy_centered:wytE4	0.38	0.59	-0.78	1.53	1.00	32285	27151

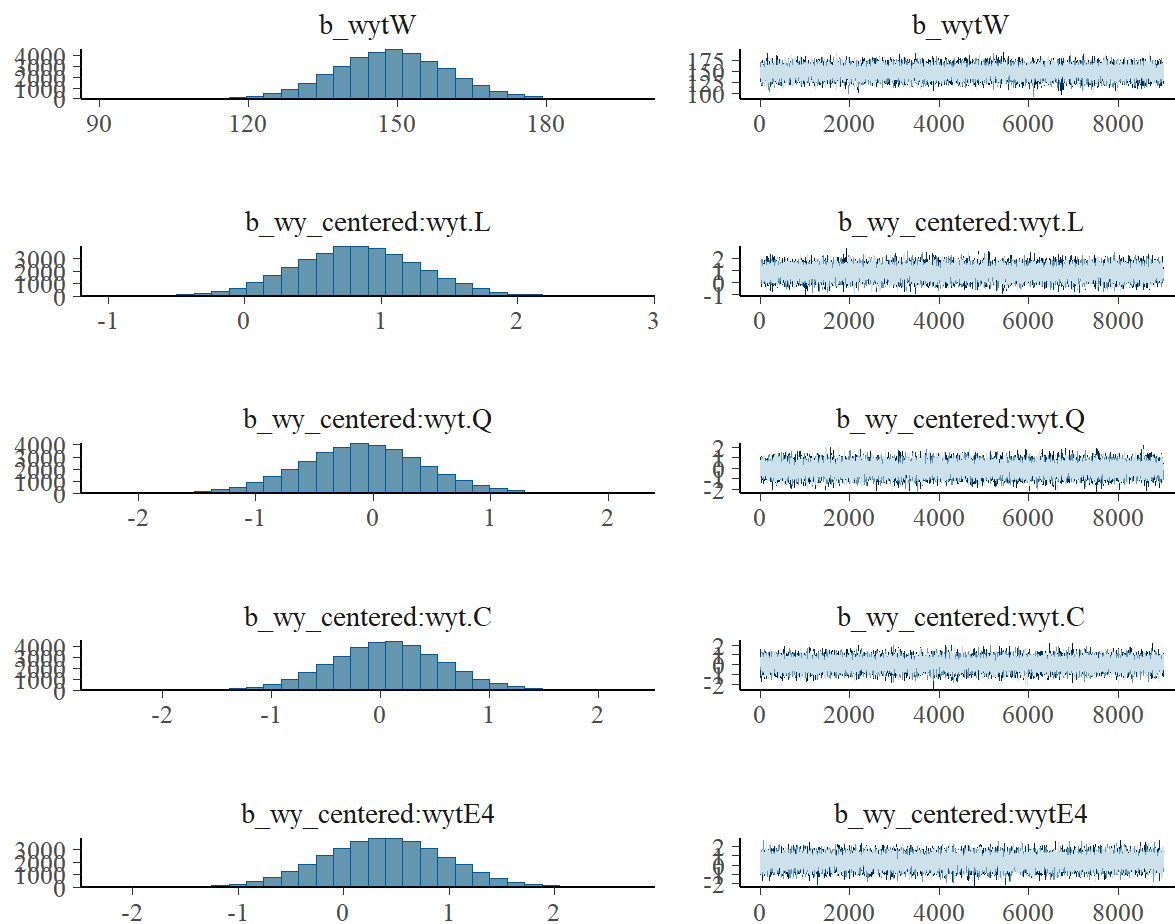
Further Distributional Parameters:

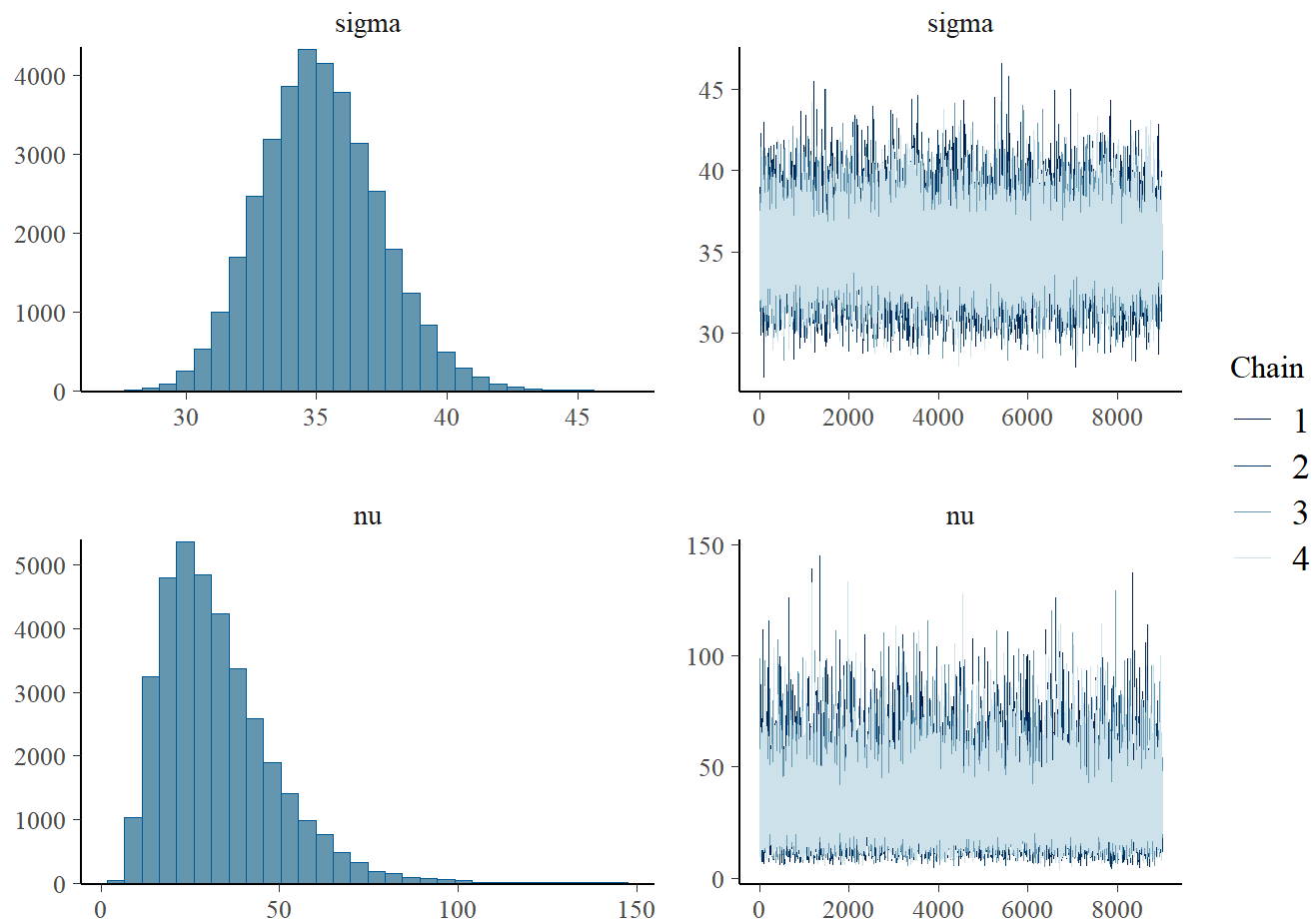
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	35.16	2.27	30.98	39.89	1.00	46290	26426
nu	32.48	15.60	11.06	70.17	1.00	49233	25595

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(m3.day10.wy.wyt)
```

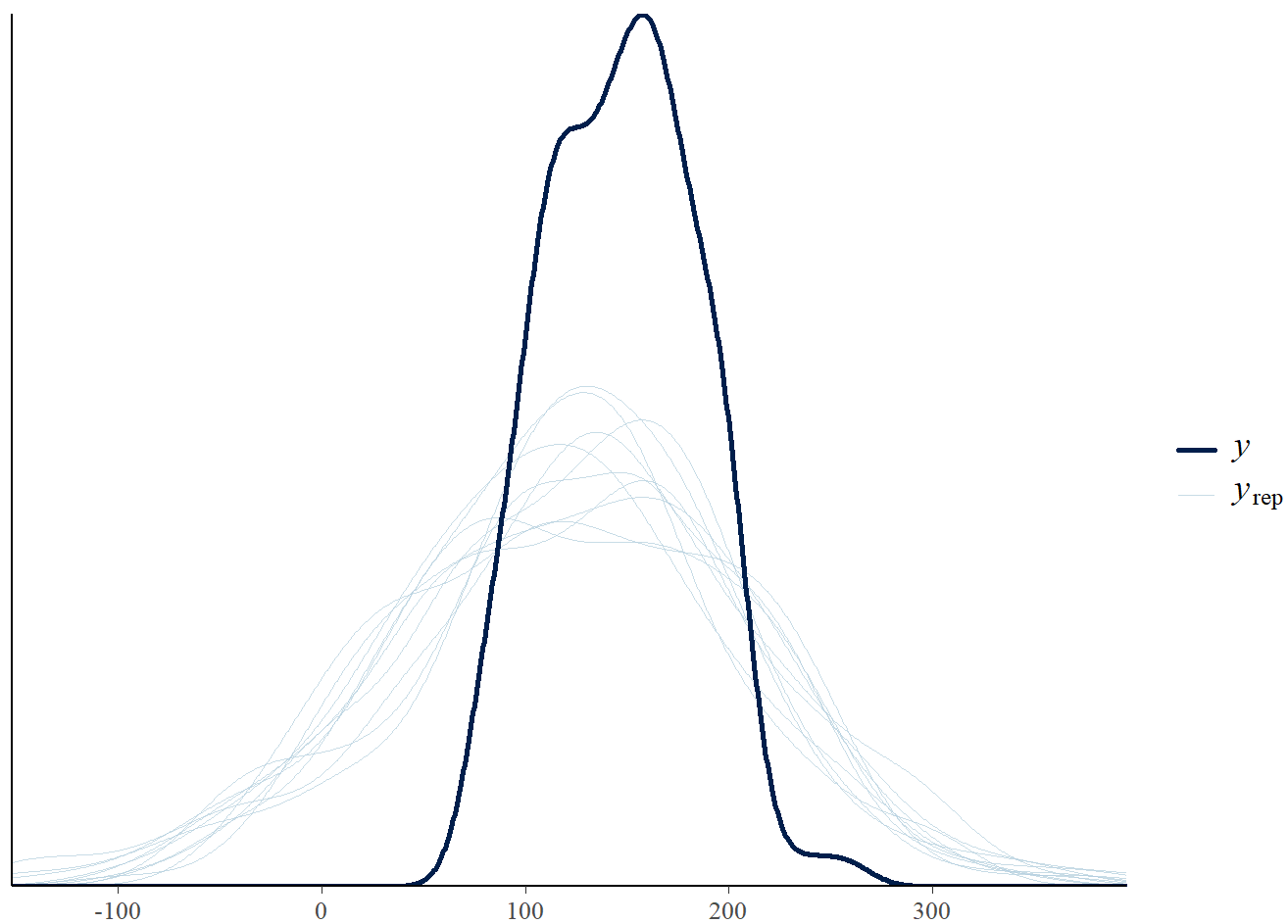






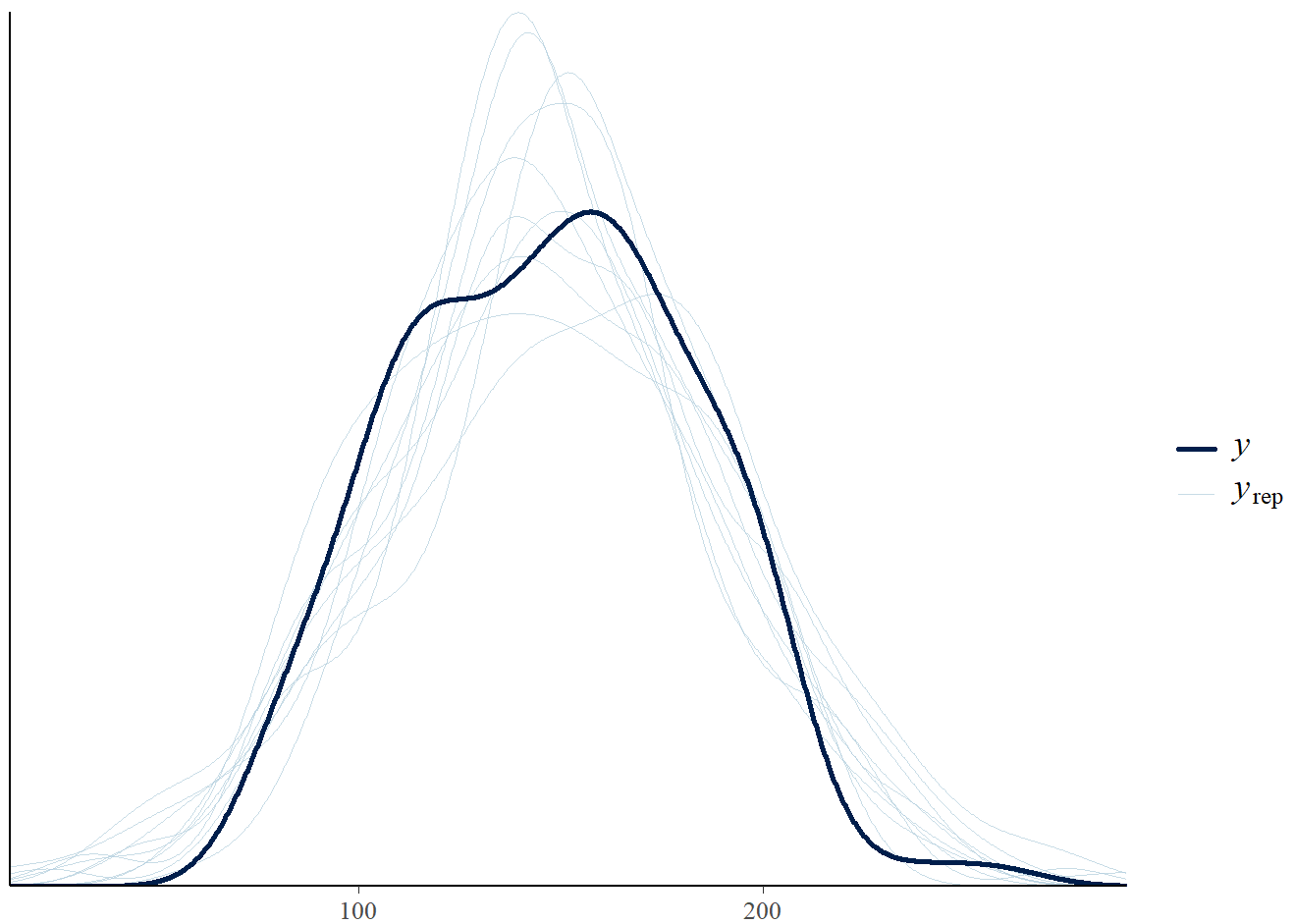
```
# posterior predictive check (see density overlay figures)
pp_check(m1.day10.wy.wyt)
```

Using 10 posterior draws for ppc type 'dens_overlay' by default.



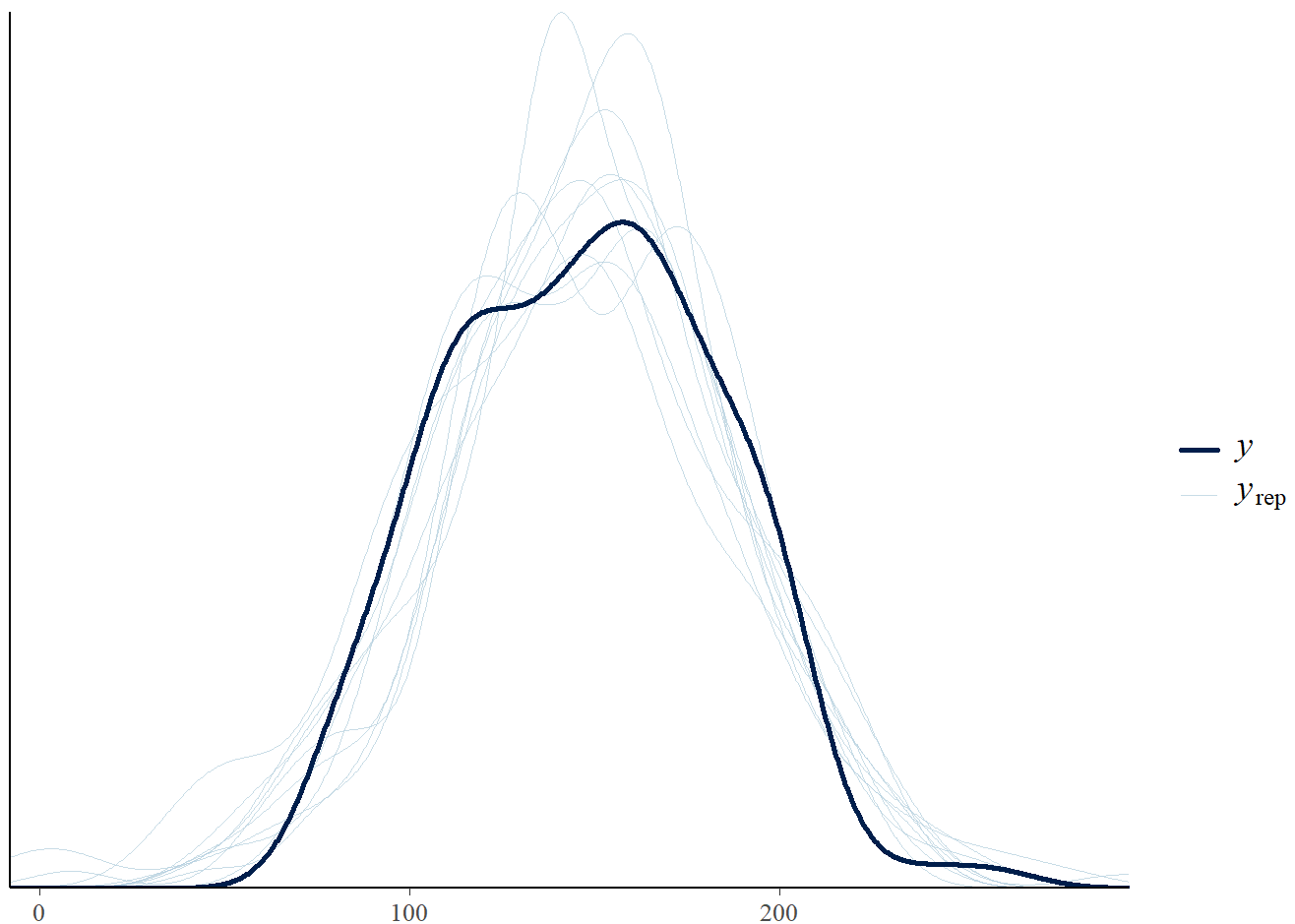
```
pp_check(m2.day10.wy.wyt)
```

Using 10 posterior draws for ppc type 'dens_overlay' by default.



```
pp_check(m3.day10.wy.wyt)
```

Using 10 posterior draws for ppc type 'dens_overlay' by default.



```
# calculate L00 for the original water_year + water_year_type Gaussian model
m1.day10.wy.wyt <- add_criterion(m1.day10.wy.wyt, criterion = "loo")

# calculate L00 for the water_year * water_year_type Gaussian model
m2.day10.wy.wyt <- add_criterion(m2.day10.wy.wyt, criterion = "loo")

# calculate L00 for the water_year * water_year_type Student-t "robust" model
m3.day10.wy.wyt <- add_criterion(m3.day10.wy.wyt, criterion = "loo")

# compare the three day10 wyt models with interaction terms
loo_compare(m1.day10.wy.wyt, m2.day10.wy.wyt, m3.day10.wy.wyt, criterion = "loo")
```

	elpd_diff	se_diff
m2.day10.wy.wyt	0.0	0.0
m3.day10.wy.wyt	-1.6	0.2
m1.day10.wy.wyt	-95.9	10.7

```
# no real difference in model performance among the 3...
```

```
m1.day10.wy.wyt$criteria$loo
```

Computed from 36000 by 144 log-likelihood matrix.

	Estimate	SE
elpd_loo	-821.0	10.8
p_loo	3.3	0.7
looic	1641.9	21.7

MCSE of elpd_loo is 0.0.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.7, 1.2]).

All Pareto k estimates are good ($k < 0.7$).

See help('pareto-k-diagnostic') for details.

```
m2.day10.wy.wyt$criteria$loo
```

Computed from 36000 by 144 log-likelihood matrix.

	Estimate	SE
elpd_loo	-725.1	6.1
p_loo	10.1	1.0
looic	1450.1	12.1

MCSE of elpd_loo is 0.0.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.1]).

All Pareto k estimates are good ($k < 0.7$).

See help('pareto-k-diagnostic') for details.

```
m3.day10.wy.wyt$criteria$loo
```

Computed from 36000 by 144 log-likelihood matrix.

	Estimate	SE
elpd_loo	-726.6	6.1
p_loo	10.5	1.0
looic	1453.3	12.3

MCSE of elpd_loo is 0.0.

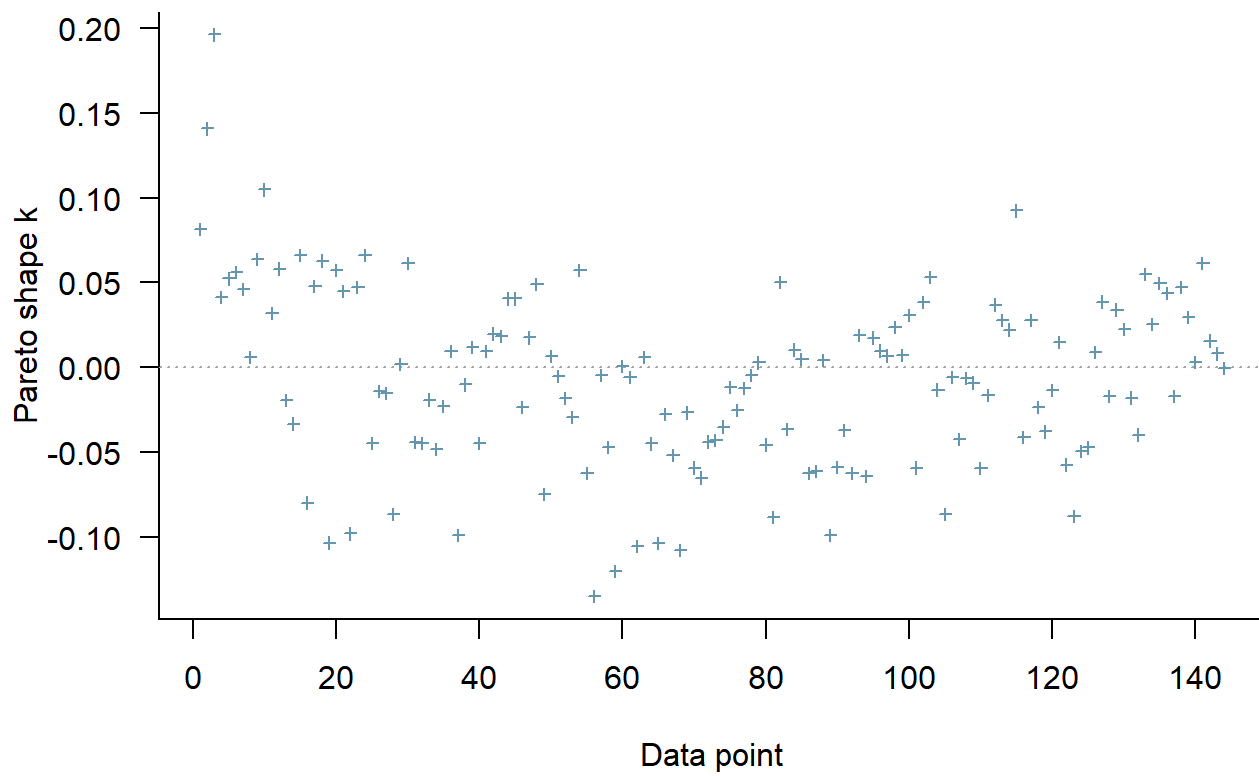
MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.1]).

All Pareto k estimates are good ($k < 0.7$).

See help('pareto-k-diagnostic') for details.

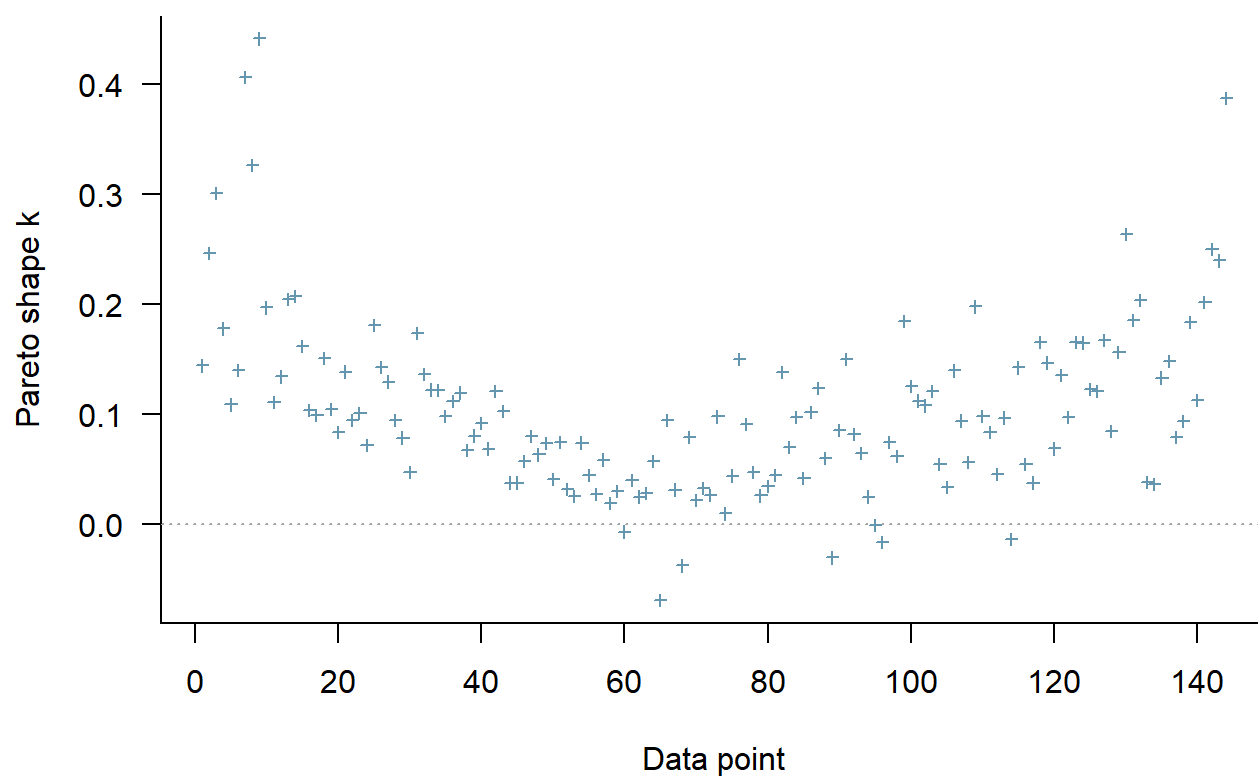
```
# let's plot the k values for each model (cause I like these figures)
plot(m1.day10.wy.wyt$criteria$loo)
```

PSIS diagnostic plot



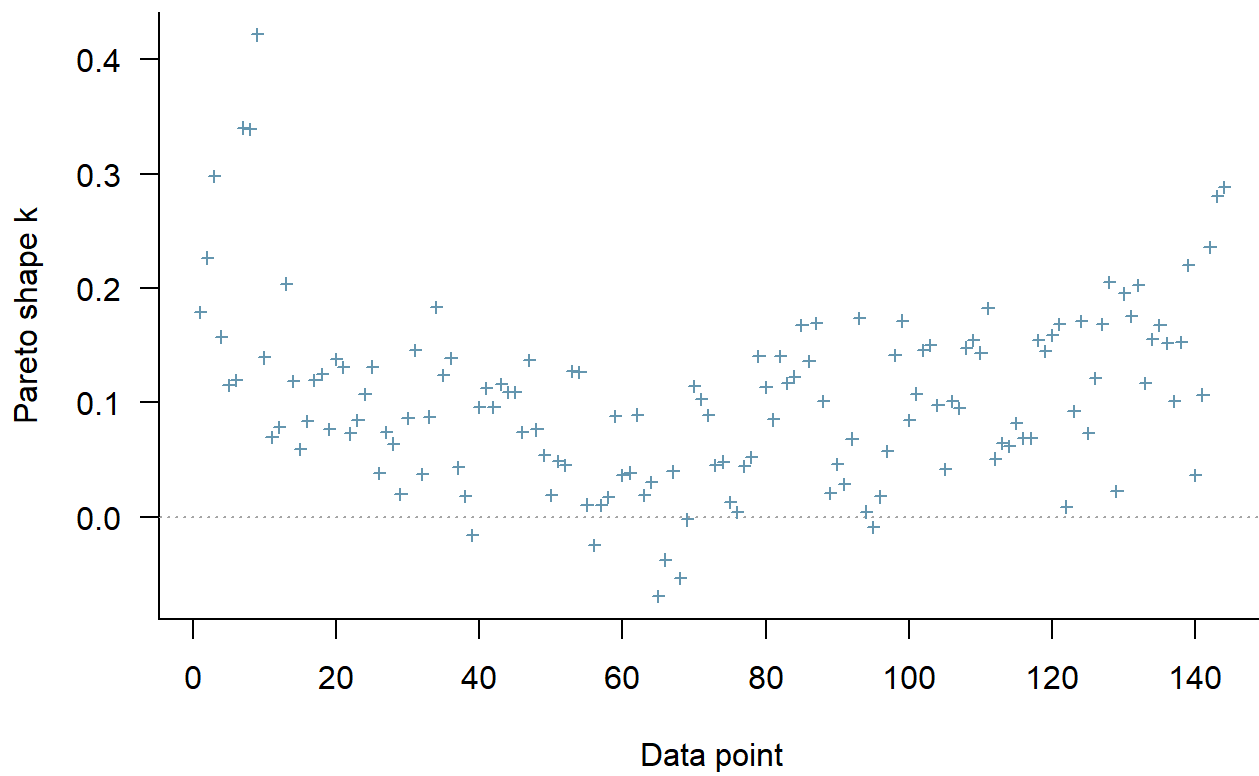
```
plot(m2.day10.wy.wyt$criteria$loo)
```

PSIS diagnostic plot



```
plot(m3.day10.wy.wyt$criteria$loo)
```

PSIS diagnostic plot



```
# everything well below 0.7...
```